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Group common downlink control information enhancements for partial frequency sounding of multiple UEs

Abstract

Aspects of the disclosure relate to the use of enhanced downlink control information to communicate partial frequency sounding configuration to user equipment (UE). A UE may receive a partial frequency sounding configuration generated for the UE the downlink control information, transmit a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration, and refrain from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE. A format for the enhanced downlink control information may be based on standards defined the downlink control information formats.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is the U.S. national stage of PCT patent application number PCT/CN2020/113385 filed on Sep. 4, 2020, and assigned to the assignee hereof and hereby expressly incorporated by reference herein as if fully set forth below in its entirety and for all applicable purposes.

INTRODUCTION

(2) The technology discussed below relates generally to wireless communication networks, and more particularly, to configuration of sounding reference signal (SRS) resources in beam-based communication scenarios.

(3) In wireless communication systems, such as those specified under standards for 5G New Radio (NR), a base station and user equipment (UE) may utilize beamforming to compensate for high path loss and short range. Beamforming is a signal processing technique used with an antenna array for directional signal transmission and/or reception. Each antenna in the antenna array transmits a signal that is combined with other signals of other antennas of the same array in such a way that signals at particular angles experience constructive interference while others experience destructive

interference.

(4) The base station and the UE can select one or more beam pair links (BPLs) for communication therebetween on the downlink and/or the uplink. Each BPL includes corresponding transmit and receive beams on the base station and UE. For example, on the uplink, a BPL includes a transmit beam on the UE and a receive beam on the base station. The base station and UE may select one or more beams forming BPLs for communication of uplink and downlink signals therebetween using a downlink beam management scheme and/or an uplink beam management scheme. In an example of uplink beam management scheme, uplink beams (e.g., an uplink BPL) may be selected by the base station based on received beamformed uplink reference signals, such as SRSs. When the channel is reciprocal, the base station may further derive the downlink beams (e.g., a downlink BPL) to communicate with the UE based on the received SRSs.

(5) The UE may transmit the SRSs in accordance with SRS resources configured by the base station. An SRS resource defines the time-frequency resource location and other parameters associated with transmission of the SRS. One or more SRS resources may be included in an SRS resource set that may be configured to enable aperiodic, semi-persistent, or periodic transmission of SRSs generated based on the SRS resources within the SRS resource set.

BRIEF SUMMARY

(6) The following presents a summary of one or more aspects of the present disclosure, in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated features of the disclosure and is intended neither to identify key or critical elements of all aspects of the disclosure nor to delineate the scope of any or all aspects of the disclosure. Its sole purpose is to present some concepts of one or more aspects of the disclosure in a form as a prelude to the more detailed description that is presented later.

(7) In one example, a method for wireless communication at a user equipment (UE) in a wireless communication network is disclosed. The method includes receiving downlink control information that includes a partial frequency sounding configuration for the UE, transmitting a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration, and refraining from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE.

(8) Another example provides a UE in a wireless communication network including a memory and a processor coupled to the memory. The processor and the memory can be configured to receive downlink control information that includes a partial frequency sounding configuration for the UE, transmit a SRS in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration, and refrain from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE.

(9) Another example provides a UE in a wireless communication network. The UE can include means for receiving downlink control information that includes a partial frequency sounding configuration for the UE, means for transmitting a SRS in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration, and means for refraining from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE.

(10) Another example provides a non-transitory computer-readable medium including code for causing one or more processors of a UE to receive downlink control information that includes a partial frequency sounding configuration for the UE, transmit a SRS in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration, and refrain from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE.

(11) In another example, a method for wireless communication at a radio access network (RAN)

entity in a wireless communication network is disclosed. The method includes configuring a SRS configuration to be used by at least one user equipment (UE), the SRS configuration including a plurality of frequency resources, generating partial frequency sounding configurations for a plurality of UEs, and transmitting the partial frequency sounding configurations in downlink control information, where each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(12) Another example provides a RAN entity in a wireless communication network including a memory and a processor coupled to the memory. The processor and the memory can be configured to configure a SRS configuration to be used by at least one UE, the SRS configuration including a plurality of frequency resources, generate partial frequency sounding configurations for a plurality of UEs, and transmit the partial frequency sounding configurations in downlink control information, where each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(13) Another example provides a RAN entity in a wireless communication network. The RAN entity can include means for configuring a SRS configuration to be used by at least one user equipment (UE), the SRS configuration including a plurality of frequency resources, means for generating partial frequency sounding configurations for a plurality of UEs, and means for transmitting the partial frequency sounding configurations in downlink control information, where each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(14) Another example provides a non-transitory computer-readable medium including code for causing one or more processors of a RAN entity to configure a SRS configuration to be used by at least one UE, the SRS configuration including a plurality of frequency resources, generate partial frequency sounding configurations for a plurality of UEs, and transmit the partial frequency sounding configurations in downlink control information, where each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(15) These and other aspects will become more fully understood upon a review of the detailed description, which follows. Other aspects, features, and examples will become apparent to those of ordinary skill in the art, upon reviewing the following description of specific, exemplary aspects of in conjunction with the accompanying figures. While features may be discussed relative to certain aspects and figures below, all aspects can include one or more of the advantageous features discussed herein. In other words, while one or more aspects may be discussed as having certain advantageous features, one or more of such features may also be used in accordance with the various aspects discussed herein. In similar fashion, while exemplary aspects may be discussed below as device, system, or method aspects, such exemplary aspects can be implemented in various devices, systems, and methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a conceptual illustration of an example of a radio access network according to some aspects.

(2) FIG. 2 is a diagram illustrating an example of a frame structure for use in a radio access network according to some aspects.

(3) FIG. 3 is a block diagram illustrating a wireless communication system supporting beamforming and/or multiple-input multiple-output (MIMO) communication according to some aspects.

- (4) FIG. 4 is a diagram illustrating an example of communication between a base station and a UE using beamforming according to some aspects.
- (5) FIGS. 5A and 5B are diagrams illustrating examples of sounding reference signal (SRS) resources according to some aspects.
- (6) FIG. 6 is a diagram illustrating exemplary SRS resource sets, each including SRS resources according to some aspects.
- (7) FIG. 7 illustrates an example of SRS requests that can be transmitted by a base station to configure SRS transmissions in a 5G NR network.
- (8) FIG. 8 illustrates resource allocation for SRS in a resource grid.
- (9) FIG. 9 illustrates a first example of a resource grid for partial frequency sounding provided in accordance with certain aspects of this disclosure.
- (10) FIG. 10 illustrates a second example of a resource grid for partial frequency sounding provided in accordance with certain aspects of this disclosure.
- (11) FIG. 11 illustrates a third example of a resource grid for partial frequency sounding provided in accordance with certain aspects of this disclosure.
- (12) FIG. 12 illustrates a fourth example of a resource grid **1200** that may be adapted according to certain aspects of this disclosure to support partial frequency sounding when multiple UEs are multiplexed using the same frequency resources.
- (13) FIG. 13 illustrates a first example in which group common downlink control information (GC-DCI) may be adapted to support configuration of partial resource sounding in accordance with certain aspects of this disclosure.
- (14) FIG. 14 illustrates a second example in which GC-DCI may be adapted to support configuration of partial resource sounding in accordance with certain aspects of this disclosure.
- (15) FIG. 15 illustrates a third example in which GC-DCI may be adapted to support configuration of partial resource sounding in accordance with certain aspects of this disclosure.
- (16) FIG. 16 is a block diagram illustrating an example of a hardware implementation for a UE employing a processing system according to some aspects.
- (17) FIG. 17 is a block diagram illustrating an example of a hardware implementation for a radio access network (RAN) entity employing a processing system according to some aspects.
- (18) FIG. 18 is a flow chart of an exemplary method for a UE to utilize a multi-TRP SRS resource set according to some aspects.
- (19) FIG. 19 is a flow chart of an exemplary method for a RAN entity to configure a multi-TRP SRS resource set according to some aspects.

DETAILED DESCRIPTION

(20) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(21) The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a “Sub-6 GHz” band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a “millimeter wave” band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a “millimeter wave”

band.

(22) With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term “sub-6 GHz” or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term “millimeter wave” or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, or may be within the EHF band.

(23) While aspects and examples are described in this application by illustration to some examples, those skilled in the art will understand that additional implementations and use cases may come about in many different arrangements and scenarios. Innovations described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects and/or uses may come about via integrated chip examples and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, AI-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described innovations may occur.

Implementations may range in spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or OEM devices or systems incorporating one or more aspects of the described innovations. In some practical settings, devices incorporating described aspects and features may also necessarily include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). It is intended that innovations described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, end-user devices, etc. of varying sizes, shapes and constitution.

(24) Various aspects of the disclosure relate to configuration of a partial frequency sounding configuration that may be used when multiple UEs are multiplexed using the same frequency resources. For example, the partial frequency sounding configuration may identify frequency resources to be used by each UE for transmitting a sounding reference signal (SRS). A SRS configuration to be used by multiple UEs may define, configure or provision a set of frequency resources to be used for SRS. The set of frequency resources may be apportioned among the multiplexed UEs. The assignment of frequency resources to the UEs may be communicated in a partial frequency sounding configuration transmitted to each UE in downlink control information (DCI). For example, the partial frequency sounding configuration may define one or more frequency resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(25) The partial frequency sounding configuration may identify allocations of frequency resources to corresponding UEs through bitmaps or indices provided in the partial frequency sounding configuration. An index in a partial frequency sounding configuration may identify one or more resources in a corresponding preconfigured table.

(26) A partial frequency sounding configuration may be transmitted in a block of the downlink control information identified with a corresponding UE. A partial frequency sounding configuration may be transmitted in downlink control information that has a format that is based on a group common downlink control information (GC-DCI) defined for use in a 5G NR network. The format of the downlink control information may be based on GC-DCI 2_3 Type A or GC-DCI 2_3 Type B. The downlink control information may be enhanced to include a partial frequency sounding field that includes a bitmap or index used to identify the one or more resources. The downlink control information may be enhanced to include a component carrier index used to indicate the component carrier set associated with the partial frequency sounding configuration.

(27) The various concepts presented throughout this disclosure may be implemented across a broad variety of telecommunication systems, network architectures, and communication standards. Referring now to FIG. 1, as an illustrative example without limitation, a schematic illustration of a radio access network **100** is provided. The RAN **100** may implement any suitable wireless communication technology or technologies to provide radio access. As one example, the RAN **100** may operate according to 3.sup.rd Generation Partnership Project (3GPP) New Radio (NR) specifications, often referred to as 5G. As another example, the RAN **100** may operate under a hybrid of 5G NR and Evolved Universal Terrestrial Radio Access Network (eUTRAN) standards, often referred to as LTE. The 3GPP refers to this hybrid RAN as a next-generation RAN, or NG-RAN. Of course, many other examples may be utilized within the scope of the present disclosure.

(28) The geographic region covered by the radio access network **100** may be divided into a number of cellular regions (cells) that can be uniquely identified by a user equipment (UE) based on an identification broadcasted over a geographical area from one access point or base station. FIG. 1 illustrates macrocells **102**, **104**, **106**, **142**, and **146**, and a small cell **108**, each of which may include one or more sectors (not shown). A sector is a sub-area of a cell. All sectors within one cell are served by the same base station. A radio link within a sector can be identified by a single logical identification belonging to that sector. In a cell that is divided into sectors, the multiple sectors within a cell can be formed by groups of antennas with each antenna responsible for communication with UEs in a portion of the cell.

(29) In general, a respective base station (BS) serves each cell. Broadly, a base station is a network element or entity in a radio access network responsible for radio transmission and reception in one or more cells to or from a UE. A BS may also be referred to by those skilled in the art as a base transceiver station (BTS), a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), an access point (AP), a Node B (NB), an eNode B (eNB), a gNode B (gNB), a transmission and reception point (TRP), or some other suitable terminology.

(30) In FIG. 1, three base stations **110**, **112**, and **114** are shown in cells **102**, **104**, and **106**, respectively; and a fourth base station **116** is shown controlling remote radio heads (RRHs) **144** and **148** in cells **142** and **146**. That is, a base station can have an integrated antenna or can be connected to an antenna or RRH by feeder cables. In the illustrated example, the cells **102**, **104**, **106**, **142**, and **146** may be referred to as macrocells, as the base stations **110**, **112**, **114**, and **116** support cells having a large size. Further, a base station **118** is shown in the small cell **108** (e.g., a microcell, picocell, femtocell, home base station, home Node B, home eNode B, etc.) which may overlap with one or more macrocells. In this example, the cell **108** may be referred to as a small cell, as the base station **118** supports a cell having a relatively small size. Cell sizing can be done according to system design as well as component constraints. It is to be understood that the radio access network **100** may include any number of wireless base stations and cells. Further, a relay node may be deployed to extend the size or coverage area of a given cell. The base stations **110**, **112**, **114**, **116**, and **118** provide wireless access points to a core network for any number of mobile apparatuses.

(31) FIG. 1 further includes an unmanned aerial vehicle (UAV) **120**, such as a quadcopter or drone, which may be configured to function as a base station. That is, in some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a mobile base station such as the UAV **120**.

(32) In general, base stations may include a backhaul interface for communication with a backhaul portion (not shown) of the network. The backhaul may provide a link between a base station and a core network (not shown), and in some examples, the backhaul may provide interconnection between the respective base stations. The core network may be a part of a wireless communication system and may be independent of the radio access technology used in the radio access network. Various types of backhaul interfaces may be employed, such as a direct physical connection, a virtual network, or the like using any suitable transport network.

(33) The RAN **100** is illustrated supporting wireless communication for multiple mobile apparatuses. A mobile apparatus is commonly referred to as user equipment (UE) in standards and specifications promulgated by the 3rd Generation Partnership Project (3GPP), but may also be referred to by those skilled in the art as a mobile station (MS), a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal (AT), a mobile terminal, a wireless terminal, a remote terminal, a handset, a terminal, a user agent, a mobile client, a client, or some other suitable terminology. A UE may be an apparatus that provides a user with access to network services.

(34) Within the present document, a “mobile” apparatus need not necessarily have a capability to move, and may be stationary. The term mobile apparatus or mobile device broadly refers to a diverse array of devices and technologies. For example, some non-limiting examples of a mobile apparatus include a mobile, a cellular (cell) phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal computer (PC), a notebook, a netbook, a smartbook, a tablet, a personal digital assistant (PDA), and a broad array of embedded systems, e.g., corresponding to an “Internet of things” (IoT). A mobile apparatus may additionally be an automotive or other transportation vehicle, a remote sensor or actuator, a robot or robotics device, a satellite radio, a global positioning system (GPS) device, an object tracking device, a drone, a multi-copter, a quad-copter, a remote control device, a consumer and/or wearable device, such as eyewear, a wearable camera, a virtual reality device, a smart watch, a health or fitness tracker, a digital audio player (e.g., MP3 player), a camera, a game console, etc. A mobile apparatus may additionally be a digital home or smart home device such as a home audio, video, and/or multimedia device, an appliance, a vending machine, intelligent lighting, a home security system, a smart meter, etc. A mobile apparatus may additionally be a smart energy device, a security device, a solar panel or solar array, a municipal infrastructure device controlling electric power (e.g., a smart grid), lighting, water, etc., an industrial automation and enterprise device, a logistics controller, agricultural equipment, etc. Still further, a mobile apparatus may provide for connected medicine or telemedicine support, i.e., health care at a distance. Telehealth devices may include telehealth monitoring devices and telehealth administration devices, whose communication may be prioritized access over other types of information, e.g., in terms of prioritized access for transport of critical service data, and/or relevant QoS for transport of critical service data.

(35) Within the RAN **100**, the cells may include UEs that may be in communication with one or more sectors of each cell. For example, UEs **122** and **124** may be in communication with base station **110**; UEs **126** and **128** may be in communication with base station **112**; UEs **130** and **132** may be in communication with base station **114**; UE **134** may be in communication with base station **118**; UEs **138** and **140** may be in communication with base station **116** via one or more of the RRHs **144** and **148**; and UE **136** may be in communication with mobile base station **120**. Here, each base station **110**, **112**, **114**, **116**, **118**, and **120** may be configured to provide an access point to a core network (not shown) for all the UEs in the respective cells. In another example, a mobile network node (e.g., UAV **120**) may be configured to function as a UE. For example, the UAV **120** may operate within cell **102** by communicating with base station **110**.

(36) In some examples, access to the air interface may be scheduled, wherein a scheduling entity (e.g., a base station **112**) allocates resources for communication among some or all devices and equipment within its service area or cell. Within the present disclosure, as discussed further below, the scheduling entity may be responsible for scheduling, assigning, reconfiguring, and releasing resources for one or more scheduled entities. That is, for scheduled communication, UEs (e.g., UE **126**), which may be scheduled entities, may utilize resources allocated by the scheduling entity **112**.

(37) Base stations are not the only entities that may function as scheduling entities. That is, in some examples, a UE may function as a scheduling entity, scheduling resources for one or more

scheduled entities (e.g., one or more other UEs). And as discussed more below, UEs may communicate directly with other UEs in peer-to-peer (P2P) fashion and/or in relay configuration. (38) In a further aspect of the RAN **100**, sidelink signals may be used between UEs without necessarily relying on scheduling or control information from a base station. For example, two or more UEs (e.g., UEs **138** and **140**) may communicate with each other using peer to peer (P2P) or sidelink signals **137** without relaying that communication through a base station (e.g., base station **144**). In some examples, the sidelink signals **137** include sidelink traffic and sidelink control. In some examples, the UEs **138** and **140** may each function as a scheduling entity or an initiating (e.g., transmitting) sidelink device and/or a scheduled entity or a receiving sidelink device. For example, the UEs **138** and **140** may function as scheduling entities or scheduled entities in a P2P network, a device-to-device (D2D), vehicle-to-vehicle (V2V) network, a vehicle-to-everything (V2X), a mesh network, or other suitable network.

(39) In the RAN **100**, the ability for a UE to communicate while moving, independent of their location, is referred to as mobility. The various physical channels between the UE and the RAN are generally set up, maintained, and released under the control of an access and mobility management function (AMF), which may include a security context management function (SCMF) that manages the security context for both the control plane and the user plane functionality and a security anchor function (SEAF) that performs authentication. In some examples, during a call with a scheduling entity, or at any other time, a UE may monitor various parameters of the signal from its serving cell as well as various parameters of neighboring cells. Depending on the quality of these parameters, the UE may maintain communication with one or more of the neighboring cells. During this time, if the UE moves from one cell to another, or if signal quality from a neighboring cell exceeds that from the serving cell for a given amount of time, the UE may undertake a handoff or handover from the serving cell to the neighboring (target) cell. For example, UE **124** may move from the geographic area corresponding to its serving cell **102** to the geographic area corresponding to a neighbor cell **106**. When the signal strength or quality from the neighbor cell **106** exceeds that of its serving cell **102** for a given amount of time, the UE **124** may transmit a reporting message to its serving base station **110** indicating this condition. In response, the UE **124** may receive a handover command, and the UE may undergo a handover to the cell **106**.

(40) Wireless communication between a RAN **100** and a UE (e.g., UE **122** or **124**) may be described as utilizing an air interface. Transmissions over the air interface from a base station (e.g., base station **110**) to one or more UEs (e.g., UE **122** and **124**) may be referred to as downlink (DL) transmission. In accordance with certain aspects of the present disclosure, the term downlink may refer to a point-to-multipoint transmission originating at a scheduling entity (described further below; e.g., base station **110**). Another way to describe this scheme may be to use the term broadcast channel multiplexing. Transmissions from a UE (e.g., UE **122**) to a base station (e.g., base station **110**) may be referred to as uplink (UL) transmissions. In accordance with further aspects of the present disclosure, the term uplink may refer to a point-to-point transmission originating at a scheduled entity (described further below; e.g., UE **122**).

(41) For example, DL transmissions may include unicast or broadcast transmissions of control information and/or data (e.g., user data traffic or other type of traffic) from a base station (e.g., base station **110**) to one or more UEs (e.g., UEs **122** and **124**), while UL transmissions may include transmissions of control information and/or traffic information originating at a UE (e.g., UE **122**). In addition, the uplink and/or downlink control information and/or traffic information may be time-divided into frames, subframes, slots, and/or symbols. As used herein, a symbol may refer to a unit of time that, in an orthogonal frequency division multiplexed (OFDM) waveform, carries one resource element (RE) per sub-carrier. A slot may carry 7 or 14 OFDM symbols. A subframe may refer to a duration of 1 ms. Multiple subframes or slots may be grouped together to form a single frame or radio frame. Of course, these definitions are not required, and any suitable scheme for organizing waveforms may be utilized, and various time divisions of the waveform may have any

suitable duration.

(42) The air interface in the RAN **100** may utilize one or more multiplexing and multiple access algorithms to enable simultaneous communication of the various devices. For example, 5G NR specifications provide multiple access for UL or reverse link transmissions from UEs **122** and **124** to base station **110**, and for multiplexing DL or forward link transmissions from the base station **110** to UEs **122** and **124** utilizing orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP). In addition, for UL transmissions, 5G NR specifications provide support for discrete Fourier transform-spread-OFDM (DFT-s-OFDM) with a CP (also referred to as single-carrier FDMA (SC-FDMA)). However, within the scope of the present disclosure, multiplexing and multiple access are not limited to the above schemes, and may be provided utilizing time division multiple access (TDMA), code division multiple access (CDMA), frequency division multiple access (FDMA), sparse code multiple access (SCMA), resource spread multiple access (RSMA), or other suitable multiple access schemes. Further, multiplexing DL transmissions from the base station **110** to UEs **122** and **124** may be provided utilizing time division multiplexing (TDM), code division multiplexing (CDM), frequency division multiplexing (FDM), orthogonal frequency division multiplexing (OFDM), sparse code multiplexing (SCM), or other suitable multiplexing schemes.

(43) Further, the air interface in the RAN **100** may utilize one or more duplexing algorithms. Duplex refers to a point-to-point communication link where both endpoints can communicate with one another in both directions. Full-duplex means both endpoints can simultaneously communicate with one another. Half-duplex means only one endpoint can send information to the other at a time. Half-duplex emulation is frequently implemented for wireless links utilizing time division duplex (TDD). In TDD, transmissions in different directions on a given channel are separated from one another using time division multiplexing. That is, at some times the channel is dedicated for transmissions in one direction, while at other times the channel is dedicated for transmissions in the other direction, where the direction may change very rapidly, e.g., several times per slot. In a wireless link, a full-duplex channel generally relies on physical isolation of a transmitter and receiver, and suitable interference cancellation technologies. Full-duplex emulation is frequently implemented for wireless links by utilizing frequency division duplex (FDD) or spatial division duplex (SDD). In FDD, transmissions in different directions may operate at different carrier frequencies (e.g., within paired spectrum). In SDD, transmissions in different directions on a given channel are separated from one another using spatial division multiplexing (SDM). In other examples, full-duplex communication may be implemented within unpaired spectrum (e.g., within a single carrier bandwidth), where transmissions in different directions occur within different sub-bands of the carrier bandwidth. This type of full-duplex communication may be referred to herein as sub-band full duplex (SBFD), also known as flexible duplex.

(44) In various implementations, the air interface in the RAN **100** may utilize licensed spectrum, unlicensed spectrum, or shared spectrum. Licensed spectrum provides for exclusive use of a portion of the spectrum, generally by virtue of a mobile network operator purchasing a license from a government regulatory body. Unlicensed spectrum provides for shared use of a portion of the spectrum without need for a government-granted license. While compliance with some technical rules is generally still required to access unlicensed spectrum, generally, any operator or device may gain access. Shared spectrum may fall between licensed and unlicensed spectrum, wherein technical rules or limitations may be required to access the spectrum, but the spectrum may still be shared by multiple operators and/or multiple RATs. For example, the holder of a license for a portion of licensed spectrum may provide licensed shared access (LSA) to share that spectrum with other parties, e.g., with suitable licensee-determined conditions to gain access.

(45) In some examples, a UE (e.g., UE **138**) may be in the coverage area of more than one cell (e.g., cells **142** and **146**). In this example, each RRH **144** and **148** serving one of the cells **142** and **146** may function as a transmission and reception point (TRP) in a coordinated multi-point

(COMP) network configuration in which downlink and/or uplink signals may be transmitted between the UE **138** and each of the TRPs **144** and **148**. For example, downlink data may be simultaneously transmitted to the UE **138** from each of the TRPs **144** and **148** to reduce interference, increase the data rate, and/or increase the received power. As another example, downlink signals may be transmitted from one TRP (e.g., TRP **144**) and uplink signals may be received at another TRP (e.g., TRP **148**). In some examples, the TRPs **144** and **148** may be configured using a centralized RAN architecture in which base station **116** operates to coordinate transmissions and receptions between the UE **138** and TRPs **144** and **148**.

(46) In addition, beamformed signals may be utilized between the UE **138** and each of the TRPs **144** and **148** communicating, for example, over a mmWave carrier. To facilitate transmission of uplink signals from the UE to one or both of the TRPs using uplink beams, the base station **116** may coordinate an uplink beam management scheme in which the UE **138** may transmit uplink reference signals, such as sounding reference signals (SRSs) to each of the TRPs **144** and **148**. Based on the SRS measurements, the base station **116** may select one or more uplink beams for the UE **138** to transmit uplink signals to one or more of the TRPs **144** and **148**. The UE **138** may transmit the SRSs in accordance with SRS resources configured by the base station **116**. An SRS resource defines the time-frequency resource location and other parameters associated with transmission of the SRS.

(47) To simplify the configuration of SRS resources across multiple TRPs **144** and **148**, in various aspects of the disclosure, the base station **116** may include an SRS manager **150** for configuring an SRS resource set including SRS resources associated with each of the TRPs **144** and **148**. The base station **116** may then transmit an SRS configuration of the SRS resource set to the UE **138** via, for example, TRP **144**. In addition, the UE **138** may further include an SRS manager **152** configured to use the SRS configuration to generate and transmit a plurality of SRSs to the multiple TRPs **144** and **148**. For example, each of the TRPs **144** and **148** may receive at least one of the SRSs transmitted by the UE **138** in accordance with the SRS configuration.

(48) Various aspects of the present disclosure will be described with reference to an OFDM waveform, schematically illustrated in FIG. 2. It should be understood by those of ordinary skill in the art that the various aspects of the present disclosure may be applied to an SC-FDMA waveform in substantially the same way as described herein below. That is, while some examples of the present disclosure may focus on an OFDM link for clarity, it should be understood that the same principles may be applied as well to SC-FDMA waveforms.

(49) Referring now to FIG. 2, an expanded view **200** of an exemplary DL subframe **202** is illustrated, showing an OFDM resource grid. However, as those skilled in the art will readily appreciate, the PHY transmission structure for any particular application may vary from the example described here, depending on any number of factors. Here, time is in the horizontal direction with units of OFDM symbols; and frequency is in the vertical direction with units of subcarriers.

(50) The resource grid **204** may be used to schematically represent time-frequency resources for a given antenna port. That is, in a multiple-input-multiple-output (MIMO) implementation with multiple antenna ports available, a corresponding multiple number of resource grids **204** may be available for communication. The resource grid **204** is divided into multiple resource elements (REs) **206**. An RE, which is 1 subcarrier $\times$ 1 symbol, is the smallest discrete part of the time-frequency grid, and contains a single complex value representing data from a physical channel or signal. Depending on the modulation utilized in a particular implementation, each RE may represent one or more bits of information. In some examples, a block of REs may be referred to as a physical resource block (PRB) or a resource block (RB), which contains any suitable number of consecutive subcarriers in the frequency domain. In one example, an RB may include 12 subcarriers, a number independent of the numerology used. In some examples, depending on the numerology, an RB may include any suitable number of consecutive OFDM symbols in the time

domain. Within the present disclosure, it is assumed that a single RB such as the RB **208** entirely corresponds to a single direction of communication (either transmission or reception for a given device).

(51) Scheduling of UEs or sidelink devices (hereinafter collectively referred to as UEs) for downlink, uplink, or sidelink transmissions typically involves scheduling one or more resource elements **206** within one or more sub-bands or bandwidth parts (BWPs). Thus, a UE generally utilizes only a subset of the resource grid **204**. In some examples, an RB may be the smallest unit of resources that can be allocated to a UE. Thus, the more RBs scheduled for a UE, and the higher the modulation scheme chosen for the air interface, the higher the data rate for the UE. The RBs may be scheduled by a base station (e.g., gNB, eNB, etc.) or may be self-scheduled by a UE/sidelink device implementing D2D sidelink communication.

(52) Scheduling of the resources (e.g., REs **206**/RBs **208**) to transmit control and/or traffic information may be performed in a dynamic manner or a semi-persistent manner. For example, the scheduling entity (e.g., base station) may dynamically allocate a set of REs **206**/RBs **208** for the transmission of downlink control and/or data to the UE or for the transmission of uplink control and/or data from the UE. The base station may further semi-persistently allocate a set of REs **206**/RBs **208** for periodic downlink or uplink transmissions. Generally, semi-persistent scheduling (SPS) may be used for periodic communications based on defined settings. For example, SPS may be suitable for applications with small, predictable, and/or periodic payloads, such as voice over Internet protocol (VOIP) applications. On the uplink, an SPS resource may be referred to as a configured grant (CG). With CGs, scheduling information corresponding to the uplink CG may be signaled just once to the UE. Subsequently, without needing to receive additional scheduling information, the UE may periodically utilize the resources allocated in the uplink CG. The periodicity with which the UE may transmit user data traffic via the semi-persistently scheduled resources may be established when the CG is initially configured.

(53) In this illustration, the RB **208** is shown as occupying less than the entire bandwidth of the subframe **202**, with some subcarriers illustrated above and below the RB **208**. In a given implementation, the subframe **202** may have a bandwidth corresponding to any number of one or more RBs **208**. Further, in this illustration, the RB **208** is shown as occupying less than the entire duration of the subframe **202**, although this is merely one possible example.

(54) Each 1 ms subframe **202** may consist of one or multiple adjacent slots. In the example shown in FIG. 2, one subframe **202** includes four slots **210**, as an illustrative example. In some examples, a slot may be defined according to a specified number of OFDM symbols with a given cyclic prefix (CP) length. For example, a slot may include 7 or 14 OFDM symbols with a nominal CP. Additional examples may include mini-slots, sometimes referred to as shortened transmission time intervals (TTIs), having a shorter duration (e.g., one to three OFDM symbols). These mini-slots or shortened transmission time intervals (TTIs) may in some cases be transmitted occupying resources scheduled for ongoing slot transmissions for the same or for different UEs. Any number of resource blocks may be utilized within a subframe or slot.

(55) An expanded view of one of the slots **210** illustrates the slot **210** including a control region **212** and a data region **214**. In general, the control region **212** may carry control channels, and the data region **214** may carry data channels. In the example shown in FIG. 2, the control region **212** may include downlink control information and the data region **214** may include downlink data channels or uplink data channels. In addition, the slot **210** further includes an uplink sounding reference signal (SRS) **216** shown transmitted at an end of the slot **210**. For example, the SRS **216** may be transmitted over one or more symbols within the last six symbols of the slot **210**. Of course, a slot may contain all DL, all UL, or at least one DL portion and at least one UL portion. The structure illustrated in FIG. 2 is merely exemplary in nature, and different slot structures may be utilized, and may include one or more of each of the control region(s) and data region(s).

(56) Although not illustrated in FIG. 2, the various REs **206** within an RB **208** may be scheduled to

carry one or more physical channels, including control channels, shared channels, data channels, etc. Other REs **206** within the RB **208** may also carry pilots or reference signals. These pilots or reference signals may provide for a receiving device to perform channel estimation of the corresponding channel, which may enable coherent demodulation/detection of the control and/or data channels within the RB **208**.

(57) In some examples, the slot **210** may be utilized for broadcast or unicast communication. For example, a broadcast, multicast, or groupcast communication may refer to a point-to-multipoint transmission by one device (e.g., a base station, UE, or other similar device) to other devices. Here, a broadcast communication is delivered to all devices, whereas a multicast communication is delivered to multiple intended recipient devices. A unicast communication may refer to a point-to-point transmission by a one device to a single other device.

(58) In an example of cellular communication over a cellular carrier via a Uu interface, for a DL transmission, the scheduling entity (e.g., a base station) may allocate one or more REs **206** (e.g., within the control region **212**) to carry DL control information including one or more DL control channels, such as a physical downlink control channel (PDCCH), to one or more scheduled entities (e.g., UEs). The PDCCH carries downlink control information (DCI) including but not limited to power control commands (e.g., one or more open loop power control parameters and/or one or more closed loop power control parameters), scheduling information, a grant, and/or an assignment of REs for DL and UL transmissions. The PDCCH may further carry HARQ feedback transmissions such as an acknowledgment (ACK) or negative acknowledgment (NACK). HARQ is a technique well-known to those of ordinary skill in the art, wherein the integrity of packet transmissions may be checked at the receiving side for accuracy, e.g., utilizing any suitable integrity checking mechanism, such as a checksum or a cyclic redundancy check (CRC). If the integrity of the transmission is confirmed, an ACK may be transmitted, whereas if not confirmed, a NACK may be transmitted. In response to a NACK, the transmitting device may send a HARQ retransmission, which may implement chase combining, incremental redundancy, etc.

(59) A base station (BS), such as the BS **110** in the wireless communication network **100** illustrated in FIG. 1, can send information to a user equipment (UE) (e.g., a UE **122**) regarding the slot format in a downlink control region. For example, the BS **110** can send the information to the UE **122** in a downlink control channel, such as the group common (GC) physical downlink control channel (PDCCH). The GC PDCCH refers to a channel, for example a PDCCH, that carries information, such as a slot format indicator (SFI) via common downlink control information (DCI), intended for a group of UEs. The UEs may be radio resource control (RRC) configured to decode the GC PDCCH. A SFI indicates the format of a current slot and/or future slot(s). The UE can use the information in the SFI to determine (identify, derive, etc.) which symbols in a slot are for uplink or downlink, or other purposes (e.g., such as sidelink, blank, or reserved).

(60) The base station may further allocate one or more REs **206** (e.g., in the control region **212** or the data region **214**) to carry other DL signals, such as a demodulation reference signal (DMRS); a phase-tracking reference signal (PT-RS); a channel state information (CSI) reference signal (CSI-RS); a primary synchronization signal (PSS); and a secondary synchronization signal (SSS). A UE may utilize the PSS and SSS to achieve radio frame, subframe, slot, and symbol synchronization in the time domain, identify the center of the channel (system) bandwidth in the frequency domain, and identify the physical cell identity (PCI) of the cell. The synchronization signals PSS and SSS, and in some examples, the PBCH and a PBCH DMRS, may be transmitted in a synchronization signal block (SSB). The PBCH may further include a master information block (MIB) that includes various system information, along with parameters for decoding a system information block (SIB). The SIB may be, for example, a SystemInformationType 1 (SIB1) that may include various additional system information. Examples of system information transmitted in the MIB may include, but are not limited to, a subcarrier spacing, system frame number, a configuration of a PDCCH control resource set (CORESET) (e.g., PDCCH CORESET0), and a search space for

SIB1. Examples of additional system information transmitted in the SIB1 may include, but are not limited to, a random access search space, downlink configuration information, and uplink configuration information. The MIB and SIB1 together provide the minimum system information (SI) for initial access.

(61) In an UL transmission, the scheduled entity (e.g., UE) may utilize one or more REs **206** (e.g., within the control region **212**, which may be at the end of the slot **210**) to carry UL control information (UCI) including one or more UL control channels, such as a physical uplink control channel (PUCCH), to the scheduling entity. UCI may include a variety of packet types and categories. In some examples, the UCI may include a scheduling request (SR), i.e., request for the scheduling entity to schedule uplink transmissions. Here, in response to the SR transmitted on the UCI, the scheduling entity may transmit downlink control information (DCI) that may schedule resources for uplink packet transmissions. UCI may also include HARQ feedback, channel state feedback (CSF), such as a CSI report, or any other suitable UCI. The scheduled entity (e.g., UE) may further utilize one or more REs **206** (e.g., within the control region **212** and/or the data region **214**) to transmit pilots, reference signals, and other information configured to enable or assist in decoding uplink data transmissions and/or in uplink beam management, such as one or more DMRSs and SRSs **216**.

(62) In addition to control information, one or more REs **206** (e.g., within the data region **214**) may be allocated for data traffic. Such data traffic may be carried on one or more traffic channels, such as, for a DL transmission, a physical downlink shared channel (PDSCH); or for an UL transmission, a physical uplink shared channel (PUSCH). In some examples, one or more REs **206** within the data region **214** may be configured to carry other signals, such as one or more SIBs and DMRSs.

(63) In an example of sidelink communication over a sidelink carrier via a PC5 interface, the control region **212** of the slot **210** may include a physical sidelink control channel (PSCCH) including sidelink control information (SCI) transmitted by an initiating (transmitting) sidelink device (e.g., V2X or other sidelink device) towards a set of one or more other receiving sidelink devices. The data region **214** of the slot **210** may include a physical sidelink shared channel (PSSCH) including sidelink data traffic transmitted by the initiating (transmitting) sidelink device within resources reserved over the sidelink carrier by the transmitting sidelink device via the SCI. Other information may further be transmitted over various REs **206** within slot **210**. For example, HARQ feedback information may be transmitted in a physical sidelink feedback channel (PSFCH) within the slot **210** from the receiving sidelink device to the transmitting sidelink device. In addition, one or more reference signals, such as a sidelink SSB and/or a sidelink CSI-RS, may be transmitted within the slot **210**.

(64) These physical channels described above are generally multiplexed and mapped to transport channels for handling at the medium access control (MAC) layer. Transport channels carry blocks of information called transport blocks (TB). The transport block size (TBS), which may correspond to a number of bits of information, may be a controlled parameter, based on the modulation and coding scheme (MCS) and the number of RBs in a given transmission.

(65) The channels or carriers described above in connection with FIGS. **1** and **2** are not necessarily all of the channels or carriers that may be utilized between a scheduling entity and scheduled entities, and those of ordinary skill in the art will recognize that other channels or carriers may be utilized in addition to those illustrated, such as other traffic, control, and feedback channels.

(66) In some aspects of the disclosure, the scheduling entity and/or scheduled entity may be configured for beamforming and/or multiple-input multiple-output (MIMO) technology. FIG. **3** illustrates an example of a wireless communication system **300** supporting beamforming and/or MIMO. In a MIMO system, a transmitter **302** includes multiple transmit antennas **304** (e.g., N transmit antennas) and a receiver **306** includes multiple receive antennas **308** (e.g., M receive antennas). Thus, there are $N \times M$ signal paths **310** from the transmit antennas **304** to the receive

antennas **308**. Each of the transmitter **302** and the receiver **306** may be implemented, for example, within a scheduling entity, a scheduled entity, or any other suitable wireless communication device. (67) The use of such multiple antenna technology enables the wireless communication system to exploit the spatial domain to support spatial multiplexing, beamforming, and transmit diversity. Spatial multiplexing may be used to transmit different streams of data, also referred to as layers, simultaneously on the same time-frequency resource. The data streams may be transmitted to a single UE to increase the data rate or to multiple UEs to increase the overall system capacity, the latter being referred to as multi-user MIMO (MU-MIMO). This is achieved by spatially precoding each data stream (i.e., multiplying the data streams with different weighting and phase shifting) and then transmitting each spatially precoded stream through multiple transmit antennas on the downlink. The spatially precoded data streams arrive at the UE(s) with different spatial signatures, which enables each of the UE(s) to recover the one or more data streams destined for that UE. On the uplink, each UE transmits a spatially precoded data stream, which enables the base station to identify the source of each spatially precoded data stream.

(68) The number of data streams or layers corresponds to the rank of the transmission. In general, the rank of the MIMO system **300** is limited by the number of transmit or receive antennas **304** or **308**, whichever is lower. In addition, the channel conditions at the UE, as well as other considerations, such as the available resources at the base station, may also affect the transmission rank. For example, the rank (and therefore, the number of data streams) assigned to a particular UE on the downlink may be determined based on the rank indicator (RI) transmitted from the UE to the base station. The RI may be determined based on the antenna configuration (e.g., the number of transmit and receive antennas) and a measured signal-to-interference-and-noise ratio (SINR) on each of the receive antennas. The RI may indicate, for example, the number of layers that may be supported under the current channel conditions. The base station may use the RI, along with resource information (e.g., the available resources and amount of data to be scheduled for the UE), to assign a transmission rank to the UE.

(69) In one example, as shown in FIG. 3, a rank-2 spatial multiplexing transmission on a 2×2 MIMO antenna configuration will transmit one data stream from each transmit antenna **304**. Each data stream reaches each receive antenna **308** along a different signal path **310**. The receiver **306** may then reconstruct the data streams using the received signals from each receive antenna **308**.

(70) Beamforming is a signal processing technique that may be used at the transmitter **302** or receiver **306** to shape or steer an antenna beam (e.g., a transmit beam or receive beam) along a spatial path between the transmitter **302** and the receiver **306**. Beamforming may be achieved by combining the signals communicated via antennas **304** or **308** (e.g., antenna elements of an antenna array) such that some of the signals experience constructive interference while others experience destructive interference. To create the desired constructive/destructive interference, the transmitter **302** or receiver **306** may apply amplitude and/or phase offsets to signals transmitted or received from the antenna elements **304** or **308** associated with the transmitter **302** or receiver **306**. In some examples, the antenna elements may be mapped to antenna ports for generation of beams. Here, the term antenna port refers to a logical port (e.g., a beam) over which a signal (e.g., a data stream or layer) may be transmitted. In an example of a base station, an antenna array may include 128 antenna elements (e.g., within a 16×8 array) that may be mapped to 32 antenna ports by an 8×1 combiner.

(71) To facilitate transmission of SRSs using uplink beams from, for example, a transmitter **302** to a receiver **306**, each of the transmitter **302** and receiver **306** may include a respective SRS manager **312a** and **312b** configured to utilize an SRS configuration for an SRS resource set including SRS resources associated with multiple TRPs. For example, the SRS manager **312b** in the receiver **306** may be configured to generate the SRS configuration and provide the SRS configuration to the transmitter **302**. In addition, the SRS manager **312a** in the transmitter **302** may be configured to utilize the SRS configuration to generate a plurality of SRSs for transmission towards multiple

receivers **306** (one of which is shown in FIG. 3).

(72) In 5G New Radio (NR) systems, particularly for above 6 GHz or mmWave systems, beamformed signals may be utilized for most downlink channels, including the physical downlink control channel (PDCCH) and physical downlink shared channel (PDSCH). In addition, broadcast control information, such as the SSB, slot format indicator (SFI), and paging information, may be transmitted in a beam-sweeping manner to enable all scheduled entities (UEs) in the coverage area of a transmission and reception point (TRP) (e.g., a gNB) to receive the broadcast control information. In addition, for UEs configured with beamforming antenna arrays, beamformed signals may also be utilized for uplink channels, including the physical uplink control channel (PUCCH) and physical uplink shared channel (PUSCH). However, it should be understood that beamformed signals may also be utilized by enhanced mobile broadband (eMBB) gNBs for sub-6 GHz systems.

(73) A base station (e.g., gNB) may generally be capable of communicating with UEs using transmit beams (e.g., downlink transmit beams) of varying beam widths. For example, a base station may be configured to utilize a wider beam when communicating with a UE that is in motion and a narrower beam when communicating with a UE that is stationary. The UE may further be configured to utilize one or more downlink receive beams to receive signals from the base station. In some examples, to select one or more downlink transmit beams and one or more downlink receive beams for communication with a UE, the base station may transmit a reference signal, such as an SSB or CSI-RS, on each of a plurality of downlink transmit beams in a beam-sweeping manner. The UE may measure the reference signal received power (RSRP) on each of the downlink transmit beams using one or more downlink receive beams on the UE and transmit a Layer 1 (L1) measurement report to the base station indicating the RSRP of one or more of the measured downlink transmit beams. The base station may then select one or more serving downlink beams (e.g., downlink transmit beams and downlink receive beams) for communication with the UE based on the L1 measurement report. The resulting selected downlink transmit beam and downlink receive beam may form a downlink beam pair link. In other examples, when the channel is reciprocal, the base station may derive the particular downlink beam(s) to communicate with the UE based on uplink measurements of one or more uplink reference signals, such as sounding reference signals (SRSs).

(74) Similarly, uplink beams (e.g., uplink transmit beam(s) at the UE and uplink receive beam(s) at the base station) may be selected by measuring the RSRP of received uplink reference signals (e.g., SRSs) or downlink reference signals (e.g., SSBs or CSI-RSs) during an uplink or downlink beam sweep. For example, the base station may determine the uplink beams either by uplink beam management via an SRS beam sweep with measurement at the base station or by downlink beam management via an SSB/CSI-RS beam sweep with measurement at the UE. The selected uplink beam may be indicated by a selected SRS resource identifier (SRI) when implementing uplink beam management or a selected SSB/CSI-RS resource when implementing downlink beam management. For example, the selected SSB/CSI-RS resource can have a spatial relation to the selected uplink transmit beam (e.g., the uplink transmit beam utilized for the PUCCH, SRS, and/or PUSCH). The resulting selected uplink transmit beam and uplink receive beam may form an uplink beam pair link.

(75) FIG. 4 is a diagram illustrating communication between a base station **404** and a UE **402** using beamformed signals according to some aspects. The base station **404** may be any of the base stations (e.g., gNBs) or scheduling entities illustrated in FIGS. 1 and/or 2, and the UE **402** may be any of the UEs or scheduled entities illustrated in FIGS. 1 and/or 2.

(76) In the example shown in FIG. 4, the base station **404** is configured to generate a plurality of beams **406a-406h**, each associated with a different beam direction. In addition, the UE **402** is configured to generate a plurality of beams **408a-408e**, each associated with a different beam direction. The base station **404** and UE **402** may select one or more beams **406a-406h** on the base

station **404** and one or more beams **408a-408e** on the UE **402** for communication of uplink and downlink signals therebetween using a downlink beam management scheme and/or an uplink beam management scheme.

(77) In an example of a downlink beam management scheme for selection of downlink beams, the base station **404** may be configured to sweep or transmit on each of a plurality of downlink transmit beams **406a-406h** during one or more synchronization slots. For example, the base station **404** may transmit a reference signal, such as an SSB or CSI-RS, on each beam in the different beam directions during the synchronization slot. Transmission of the beam reference signals may occur periodically (e.g., as configured via radio resource control (RRC) signaling by the gNB), semi-persistently (e.g., as configured via RRC signaling and activated/deactivated via medium access control-control element (MAC-CE) signaling by the gNB), or aperiodically (e.g., as triggered by the gNB via downlink control information (DCI)). It should be noted that while some beams are illustrated as adjacent to one another, such an arrangement may be different in different aspects. For example, downlink transmit beams **406a-406h** transmitted during a same symbol may not be adjacent to one another. In some examples, the base station **404** may transmit more or less beams distributed in all directions (e.g., 360 degrees).

(78) In addition, the UE **402** is configured to receive the downlink beam reference signals on a plurality of downlink receive beams **408a-408e**. In some examples, the UE **402** searches for and identifies each of the downlink transmit beams **406a-406h** based on the beam reference signals. The UE **402** then performs beam measurements (e.g., RSRP, SINR, RSRQ, etc.) on the beam reference signals on each of the downlink receive beams **408a-408e** to determine the respective beam quality of each of the downlink transmit beams **406a-406h** as measured on each of the downlink receive beams **408a-408e**.

(79) The UE **402** can generate and transmit an L1 measurement report, including the respective beam index (beam identifier (ID)) and beam measurement of one or more of the downlink transmit beam **406a-406h** on one or more of the downlink receive beams **408a-408e** to the base station **404**. The base station **404** may then select one or more downlink transmit beams on which to transmit unicast downlink control information and/or user data traffic to the UE **402**. In some examples, the selected downlink transmit beam(s) have the highest gain from the beam measurement report. In some examples, the UE **402** can further identify the downlink transmit beams selected by the base station from the beam measurements. Transmission of the beam measurement report may occur periodically (e.g., as configured via RRC signaling by the gNB), semi-persistently (e.g., as configured via RRC signaling and activated/deactivated via MAC-CE signaling by the gNB), or aperiodically (e.g., as triggered by the gNB via DCI).

(80) The base station **404** or the UE **402** may further select a corresponding downlink receive beam on the UE **402** for each selected serving downlink transmit beam to form a respective downlink beam pair link (BPL) for each selected serving downlink transmit beam. For example, the UE **402** can utilize the beam measurements to select the corresponding downlink receive beam for each serving downlink transmit beam. In some examples, the selected downlink receive beam to pair with a particular downlink transmit beam may have the highest gain for that particular downlink transmit beam.

(81) In one example, a single downlink transmit beam (e.g., beam **406d**) on the base station **404** and a single downlink receive beam (e.g., beam **408c**) on the UE may form a single downlink BPL used for communication between the base station **404** and the UE **402**. In another example, multiple downlink transmit beams (e.g., beams **406c**, **406d**, and **406e**) on the base station **404** and a single downlink receive beam (e.g., beam **408c**) on the UE **402** may form respective downlink BPLs used for communication between the base station **404** and the UE **402**. In another example, multiple downlink transmit beams (e.g., beams **406c**, **406d**, and **406e**) on the base station **404** and multiple downlink receive beams (e.g., beams **408c** and **408d**) on the UE **402** may form multiple downlink BPLs used for communication between the base station **404** and the UE **402**. In this

example, a first downlink BPL may include downlink transmit beam **406c** and downlink receive beam **408c**, a second downlink BPL may include downlink transmit beam **408d** and downlink receive beam **408c**, and a third downlink BPL may include downlink transmit beam **408e** and downlink receive beam **408d**.

(82) When the channel is reciprocal, the above-described downlink beam management scheme may also be used to select one or more uplink BPLs for uplink communication from the UE **402** to the base station **404**. For example, the downlink BPL formed of beams **406d** and **408e** may also serve as an uplink BPL. Here, beam **408c** is utilized as an uplink transmit beam, while beam **406d** is utilized as an uplink receive beam.

(83) In an example of an uplink beam management scheme, the UE **402** may be configured to sweep or transmit on each of a plurality of uplink transmit beams **408a-408e**. For example, the UE **402** may transmit an SRS on each beam in the different beam directions. In addition, the base station **404** may be configured to receive the uplink beam reference signals on a plurality of uplink receive beams **406a-406h**. In some examples, the base station **404** searches for and identifies each of the uplink transmit beams **408a-408e** based on the beam reference signals. The base station **404** then performs beam measurements (e.g., RSRP, SINR, RSRQ, etc.) on the beam reference signals on each of the uplink receive beams **406a-406h** to determine the respective beam quality of each of the uplink transmit beams **408a-408e** as measured on each of the uplink receive beams **406a-406h**.

(84) The base station **404** may then select one or more uplink transmit beams on which the UE **402** will transmit unicast downlink control information and/or user data traffic to the base station **404**. In some examples, the selected uplink transmit beam(s) have the highest gain. The base station **404** may further select a corresponding uplink receive beam on the base station **404** for each selected serving uplink transmit beam to form a respective uplink beam pair link (BPL) for each selected serving uplink transmit beam. For example, the base station **404** can utilize the uplink beam measurements to select the corresponding uplink receive beam for each serving uplink transmit beam. In some examples, the selected uplink receive beam to pair with a particular uplink transmit beam may have the highest gain for that particular uplink transmit beam.

(85) The base station **404** may then notify the UE **402** of the selected uplink transmit beams. For example, the base station **404** may provide the SRS resource identifiers (SRIs) identifying the SRSs transmitted on the selected uplink transmit beams. In some examples, the base station **404** may apply each selected uplink transmit beam (and corresponding uplink receive beam) to an uplink signal (e.g., PUCCH, PUSCH, etc.) and transmit the respective SRIs associated with the selected uplink transmit beams applied to each uplink signal to the UE **402**. When the channel is reciprocal, the above-described uplink beam management scheme may also be used to select one or more downlink BPLs for downlink communication from the base station **404** to the UE **402**. For example, the uplink BPLs may also be utilized as downlink BPLs.

(86) To facilitate transmission of SRSs using uplink beams from the UE **402** to the base station **404**, each of the UE **402** and base station **406** may include a respective SRS manager **410** and **412**, respectively, configured to utilize an SRS configuration for an SRS resource set including SRS resources associated with multiple TRPs (e.g., the base station **404** and at least one other base station or TRP). For example, the SRS manager **412** may be configured to generate the SRS configuration and provide the SRS configuration to the UE **402**. In addition, the SRS manager **410** may be configured to utilize the SRS configuration to generate a plurality of SRSs for transmission towards the base station **404** and at least one other base station or TRP.

(87) FIGS. 5A and 5B are diagrams illustrating examples of sounding reference signal (SRS) resources **502a** and **502b** according to some aspects. The SRS resources **502a** and **502b** are illustrated as being located within a portion of a resource grid **500** including a plurality of REs **508**. In the time domain, an SRS resource can include a number of consecutive OFDM symbols $N_{\text{sub.symb}}$ selected from the set consisting of {1, 2, 4}. Thus, an SRS resource can occupy 1, 2, or 4 OFDM symbols. In the example shown in FIG. 5A, the SRS resource **502a** includes one OFDM

symbol, while in the example shown in FIG. 5B, the SRS resource **502b** includes two OFDM symbols. In addition, an SRS resource may be located within the last six symbols of a slot. Thus, the starting symbol of an SRS resource may be designated as a number of symbols offset from the last symbol of the slot. For example, the starting symbol *I.sub.offset* may be selected from the set consisting of {0, 1, 2, 3, 4, 5}. In the example shown in FIG. 5A, the starting symbol **506a** may be indicated as *I.sub.offset*=1, while in the example shown in FIG. 5B, the starting symbol **506b** may be indicates as *I.sub.offset*=2.

(88) In the frequency domain, each SRS resource **502a** and **502b** may occupy a number of RBs *m.sub.SRS*. In some examples, the number of RBs may be between $1 \leq m.sub.SRS \leq 272$. In addition, each SRS resource **502a** and **502b** may have a transmission comb structure *k.sub.TC*, such that the SRS is transmitted every *N*th subcarrier, where *N*=2 or 4. Thus, the transmission comb structure *k.sub.TC* may be selected from the set consisting of {2, 4}. In the example shown in FIG. 5A, the SRS resource **502a** has a transmission comb structure **504a** of two, while in the example shown in FIG. 5B, the SRS resource **502b** has a transmission comb structure **504b** of four. In some examples, an SRS resource (e.g., SRS resource **502b**) occupying more than one symbol may be used to transmit a single SRS or one or more repetitions of the SRS. In addition, each SRS resource **502a** and **502b** may include up to four antenna ports that may be interleaved, for example, in the frequency domain.

(89) FIG. 6 is a diagram illustrating exemplary SRS configurations **600a-600c** for SRS resource sets **602a-602c**, each including SRS resources **604a-604f** according to some aspects. An SRS resource set may include one or more SRS resources. For example, SRS resource set **602a** (SRS Resource Set 0) includes SRS resources **604a** and **604b** (SRS Resource 0.0 and SRS Resource 0.1), SRS resource set **602b** (SRS Resource Set 1) includes SRS resource **604c** (SRS Resource 1.0), and SRS resource set **602c** (SRS Resource Set 2) includes SRS resource sets **604d**, **604e**, and **604f** (SRS Resource 2.0, SRS Resource 2.1, and SRS Resource 2.2).

(90) As indicated in FIG. 6, multiple SRS resource sets **602a-602c** may be configured for a UE. In addition, each SRS resource set **602a-602c** may be configured to be periodic, aperiodic, or semi-persistent, such that each of the SRS resources within the corresponding SRS resource set are periodic, aperiodic, or semi-persistent, respectively. For example, the SRS resources **604a** and **604b** within SRS resource set **602a** may be periodic SRS resources, the SRS resource **604c** within SRS resource set **602b** may be aperiodic SRS resources, and the SRS resources **604d-604f** within SRS resource set **602c** may be semi-persistent SRS resources.

(91) Each SRS resource **604a-604f** includes a set of SRS resource parameters configuring the SRS resource. For example, the SRS resource parameters may include a set of port(s) (e.g., uplink beam), number of consecutive symbols (*N.sub.symb*), time domain allocation (*I.sub.offset*), repetition, transmission comb structure (*k.sub.TC*), bandwidth (*m.sub.SRS*), and other suitable parameters. Each SRS may further be quasi co-located (QCL'ed) with another reference signal, such as an SSB, CSI-RS, or another SRS. Thus, based on the QCL association (e.g., with an SSB beam, CSI-RS beam, or SRS beam), the SRS resource may be transmitted with the same spatial domain filter utilized for reception/transmission of the indicated reference signal (e.g., SSB beam, CSI-RS beam, or SRS beam).

(92) The respective sets of SRS resource parameters for each of the SRS resources in a particular SRS resource set collectively form the SRS resource set parameters for the SRS resource set. In addition, the SRS resource set itself may further include additional SRS resource set parameters. For example, the SRS resource set parameters for the aperiodic SRS resource set **602b** may further include an aperiodic trigger state (e.g., codepoint) for the aperiodic SRS resource set **602b** (e.g., up to three trigger states may be possible, each mapping to an aperiodic SRS resource set), a slot offset between the slot including the DCI triggering the aperiodic SRS resource and transmission of the SRS (e.g., SRS is transmitted *k* slot(s) after the slot carrying the DCI containing the trigger state), and a CSI-RS resource identifier (CRI) associated with the aperiodic SRS resource set **602b** for

precoder estimation of the aperiodic SRSs. As another example, the SRS configuration for a periodic SRS resource set **602a** or semi-persistent SRS resource set **602c** may indicate the periodicity of the SRS resources (e.g., the periodicity of transmission of SRSs). The respective SRS resource set parameters then collectively form the SRS configuration **600a-600c** of the corresponding SRS resource set **602a-602c**.

(93) Certain aspects of the disclosure provide enhanced signaling that can be used for partial frequency sounding when multiple UEs are multiplexed using the same frequency resources. The enhanced signaling relates to group common downlink control information (GC-DCI), which provides a block of control information used for signaling multiple UEs. UEs can be configured to use GC-DCI for SRS transmissions.

(94) SRS transmissions by the UE can be used by a base station for CSI acquisition and beam management. Certain 5G NR network support SRS resources that span 1, 2 or 4 adjacent symbols. Each SRS resource set contains a sets of SRS resources transmitted by one UE. Transmission of the SRS resource may be aperiodic, semi-persistent or periodic. Aperiodic SRS transmissions may be triggered or otherwise signaled using DCI. A UE may be configured or provisioned with multiple resources, which may be grouped in a SRS resource set depending on use case, where use case may be antenna switching, codebook-based, non-codebook based or beam management.

(95) DCI 2_3 can be used for power control and may include a group of TPC commands for SRS transmissions by one or more UEs. Along with a TPC command, an SRS request may also be transmitted. FIG. 7 illustrates an example of SRS requests **700** that can be transmitted by a base station to configure the use GC-DCI for SRS transmissions in 5G NR network. DCI 2_3 defines a structure which contains multiple blocks, each block relating to one or more component carriers (CCs). In networks that use carrier aggregation (CA), a TPC may be provided for each CC. CA is used in certain networks to increase the data rate for one or more UEs by defining a set that includes multiple contiguous or non-contiguous CCs. In one example, 5G NR can support up to 16 contiguous and non-contiguous CCs and can aggregate new 5G bands up to approximately 1 GHz of spectrum. Certain elements of an SRS request (0, 2 bits) may determine the CC set used for CA.

(96) Certain aspects of this disclosure provide signaling enhancements related to SRS capacity, SRS coverage, and/or partial sounding for UEs that are multiplexed across a shared frequency band. SRS capacity can determine the number of UEs that can be multiplexed when each UE uses a portion of the SRS resources, thereby operating using partial sounding.

(97) FIG. 8 illustrates a resource grid **800** for a conventional resource allocation for SRS. The illustrated resource allocation employs hopping in time such that SRS resources are provided in four groups **802, 804, 806, 808** separated by time and frequency. The resources in the four groups **802, 804, 806, 808** are allocated to a single UE that can transmit SRS in each CC and in a period **810** at the end of the resource grid **800**. The SRS hopping pattern is designed to sound the full bandwidth and is RRC configured for each UE.

(98) FIG. 9 illustrates a first example of a resource grid **900** provided in accordance with certain aspects of this disclosure to support partial frequency sounding when multiple UEs are multiplexed using the same frequency resources. In this example, SRS resources are allocated among four UEs. The SRS resources are configured with hopping in time such that four RB groups are provided as SRS resources in different time intervals within a period **910** at the end of the resource grid **900**, the RB groups being separated by time and frequency. In the illustrated example, a first RB group **912** includes four RBs **902, 904, 906, 908** divided among the four UEs. Each UE is allocated four RBs for use as SRS resources, and the relative positioning of the RBs may be maintained across all four groups. In this example, a first UE may be allocated a first RB **902** that includes the highest frequency CC in a first group of resources and in each of the other groups of SRS resources.

(99) In some implementations, the relative position of SRS resources allocated to one or more UEs may vary between groups. In some implementations, the relative position of SRS resources allocated to one or more UEs may be changed after each transmission of SRS or after a configured

number of SRS transmissions. In the latter example, an index may be assigned for each UE identifying current location within one or more groups of SRS resources, and location may be varied by incrementing or decrementing the index. For example, an index or location number may cycle through the sequence {0, 1, 2, 3, 0, 1 . . . } when four resources are defined for the corresponding group of SRS resources. In some implementations, the relative position of SRS resources allocated to one or more UEs may be changed using RRC signaling.

(100) In some implementations, each UE may be allocated of SRS resources in fewer than all of the groups. For example, sufficient SRS resources in each group may not be available to allocate a resource for each UE when five or more UEs share the RS resources.

(101) FIG. **10** illustrates a second example of a resource grid **1000** provided in accordance with certain aspects of this disclosure to support partial frequency sounding when multiple UEs are multiplexed using the same frequency resources. In this example, SRS resources are allocated among four UEs. The SRS resources are configured without hopping in time and allocated in four groups **1002**, **1004**, **1006**, **1008** of frequency-contiguous SRS resources. The four groups **1002**, **1004**, **1006**, **1008** of frequency-contiguous SRS resources are provided in the same time interval **1010** near the end of the resource grid **1000**. In the illustrated example, a first group **1002** includes four SRS resources allocated to a first UE, a second group **1004** includes four SRS resources allocated to a second UE, a third group **1006** includes four SRS resources allocated to a third UE and a fourth group **1008** includes four SRS resources allocated to a fourth UE.

(102) The relative positioning of the four groups **1002**, **1004**, **1006**, **1008** in the resource grid **1000** may be maintained between transmissions. In some implementations, the relative position of the four groups **1002**, **1004**, **1006**, **1008** within the resource grid **1000** may be changed after each transmission of SRS or after a configured number of SRS transmissions. In some implementations, different UEs may be allocated different numbers of SRS resources.

(103) FIG. **11** illustrates a third example of a resource grid **1100** provided in accordance with certain aspects of this disclosure to support partial frequency sounding when multiple UEs are multiplexed using the same frequency resources. In this example, SRS resources are allocated among two UEs. The SRS resources are configured without hopping in time and provided in four groups **1102**, **1104**, **1106**, **1108**. For each UE, the allocated SRS resources are not frequency-contiguous. The four groups **1102**, **1104**, **1106**, **1108** of SRS resources are provided in the same time interval **1110** near the end of the resource grid **1100**. In the illustrated example, a first group **1102** includes four SRS resources allocated to a first UE, a second group **1104** includes four SRS resources allocated to a second UE, a third group **1106** includes four SRS resources allocated to the first UE and a fourth group **1108** includes four SRS resources allocated to the second UE. Each UE is allocated 8 SRS resources.

(104) The relative positioning of the four groups **1102**, **1104**, **1106**, **1108** in the resource grid **1100** may be maintained between transmissions. In some implementations, the relative position of the four groups **1102**, **1104**, **1106**, **1108** within the resource grid **1100** may be changed after each transmission of SRS or after a configured number of SRS transmissions. In some implementations, different UEs may be allocated different numbers of SRS resources.

(105) FIG. **12** illustrates a fourth example of a resource grid **1200** provided in accordance with certain aspects of this disclosure to support partial frequency sounding when multiple UEs are multiplexed using the same frequency resources. In this example, SRS resources are allocated among four UEs. The SRS resources are configured with hopping in time such that four groups **1202**, **1204**, **1206**, **1208** of SRS resources are provided in different time intervals within a period **1210** at the end of the resource grid **1200**, the groups being separated by time and frequency. In the illustrated example, a first group **1202** includes four SRS resources allocated to a first UE, a second group **1204** includes four SRS resources allocated to a second UE, a third group **1206** includes four SRS resources allocated to a third UE and a fourth group **1208** includes four SRS resources allocated to a fourth UE.

(106) The relative positioning of the four groups **1202**, **1204**, **1206**, **1208** in the resource grid **1200** may be maintained between transmissions. In some implementations, the relative position of the four groups **1202**, **1204**, **1206**, **1208** within the resource grid **1200** may be changed after each transmission of SRS or after a configured number of SRS transmissions. In some implementations, different UEs may be allocated different numbers of SRS resources.

(107) Certain aspects of the disclosure provide SRS resource information to UEs that are multiplexed using the same frequency resources. The SRS resource information may include information identifying the frequencies to be used by the UEs for transmitting SRS. A single UE can sound only on the sub-band of the configured SRS frequency resources when partial frequency sounding is employed. The UE can hop between a smaller set of frequency resources when partial frequency sounding is configured (see FIGS. **9** and **12**), or can sound a single contiguous sub-band without hopping (see FIG. **10**) or non-contiguous sub-bands (see FIG. **11**).

(108) Signaling provided in accordance with certain aspects of this disclosure can inform the UE which frequency resources are to be used for partial frequency sounding, including information regarding hopping and sub-bands. In one aspect, a multiple block DCI format is defined to enable the base station to configure one or more UEs for partial frequency sounding. In another aspect, an enhanced GC-DCI format is defined to enable the base station to configure one or more UEs for partial frequency sounding.

(109) A multiple block DCI format provided in accordance with an aspect of the disclosure provides a block of information for each UE. The multiple block DCI format may also provide a bitmap that can be used by each UE to identify SRS resources assigned to the UE for partial frequency sounding.

(110) The base station may use RRC signaling to configure UEs and to configure resource blocks for SRS. Referring again to FIG. **9**, for example, the base station may define the first RB group **912** that includes four RBs **902**, **904**, **906**, **908** that are to be allocated for the use of different UEs.

(111) In one aspect, DCI may be formatted with multiple blocks, and each block may correspond to one or more UEs. Each block may include a bitmap that can be used to select sub frequency resources to be sounded. The bitmap may directly identify the sub frequency resources. In the hopping example illustrated in FIG. **9**, the first RB group **912** (also referred to as the first hop) includes 4 sub-hops which can be allocated among 4 UEs. In this example, each sub-hop includes a single RB **902**, **904**, **906**, **908**, although one or more sub-hops may include more than one of the RBs **902**, **904**, **906**, **908**. A first UE may be configured with a bitmap with the value '1000' indicating that the first UE can use the first sub-hop for sounding and that the remaining 3 sub-hops are unavailable or unused for sounding by the first UE.

(112) In the non-hopping example illustrated in FIG. **10**, the RBs allocated for SRS by the base station may be divided into four RB groups **1002**, **1004**, **1006**, **1008**, and each of the RB groups **1002**, **1004**, **1006**, **1008** can be allocated for the use of different UEs. Each of the RB groups **1002**, **1004**, **1006**, **1008** includes 4 RBs, although one or more of the RB groups **1002**, **1004**, **1006**, **1008** may include more than four RBs. A first UE may be configured with a bitmap with the value '1000' indicating that the first UE can use the sub-band that includes the first RB group **1002** for sounding and that the remaining 3 RB groups **1004**, **1006**, **1008** are unavailable or unused for sounding by the first UE.

(113) In some implementations, a bitmap provided by the DCI is valid for all hops or transmissions. In some implementations, cycling within the hops can be indicated by RRC signaling. In one example, the bitmap may be cycled in a sequence such as {1000.fwdarw.0100.fwdarw.0010.fwdarw.0001.fwdarw.1000 . . . }.

(114) In some instances, the bitmap may indicate that more than one sub-resource (sub-hop or sub-band) may be sounded. For example, a bitmap with the value '1001' indicates that the first UE can use the first and last sub-hops or the first and last sub-bands for sounding. The bitmap can support non-contiguous resource sounding including when, for example, the bit map has the value '1010' or

'0101'.

(115) In one aspect, the DCI may be formatted to include multiple blocks, where each block corresponds to one or more UEs and includes an index that indicates an entry in an RRC-configured table. The entry in the table may identify which sub-frequency resource (sub-hop or sub-band) is assigned to the corresponding UE to be sounded. In the 4-UE, 4 sub-frequency resources examples illustrated in FIGS. 9, 10 and 12, a 2-bit index is sufficient to indicate one of the four possible entries of pre-configured table. The use of an index can provide benefits over bitmaps when the sub-frequency resources are unequally split, when larger numbers of UEs are to be configured, and when it is desirable to increase the number of properties that can be referenced in the preconfigured table. The size of the index may be calculated as $\log_2(\text{number of configurations to be supported})$, where the number of configurations may correspond to the number of resources for sounding use. The use of an index requires that the table be preconfigured, typically by RRC signaling.

(116) In one aspect, an enhanced GC-DCI format may be used to inform the UE which frequency resources are to be used for partial frequency sounding. FIG. 13 illustrates a first example in which GC-DCI 2_3 Type A 1300 may be adapted to support configuration of partial resource sounding. GC-DCI Type A is used to provide triggers only for antenna switching. GC-DCI 2_3 Type A is used for wideband configuration and does not include information concerning frequency resources. GC-DCI Type A provides a block 1302, 1304, 1306 for each of N UEs or groups of UEs. Each block 1302, 1304, 1306 corresponds to a set of carrier frequencies and includes one or more different TPC entries 1310. An SRS request identifier 1308 is used to select a CC set (carrier frequency). The TPC entries 1310 includes an entry for each of the corresponding CC sets.

(117) FIG. 14 illustrates a second example in which GC-DCI 2_3 Type B 1400 may be adapted to support configuration of partial resource sounding. GC-DCI Type B is used for wideband configuration to configure certain aspects of carrier frequency triggering. GC-DCI Type B includes information concerning frequency resources. GC-DCI Type B provides a block 1402, 1404, 1406 for N carrier frequencies. Each block 1402, 1404, 1406 corresponds to a single carrier frequency and includes a TPC entry 1410 for the corresponding carrier frequency. An SRS request identifier 1408 is used to select an individual CC (carrier frequency).

(118) Both GC-DCI 2_3 type A and GC-DCI 2_3 type B are used for, or support wideband sounding. Certain aspects of this disclosure define a new field for GC-DCI to indicate sub-frequency resources to be sounded by UEs configured for partial frequency sounding. The bits of the new field can indicate explicitly the partial frequency resources to be used, or may be used as an index to a preconfigured table to indicate the partial frequency resources to be used.

(119) FIG. 13 illustrates a first example of enhanced GC-DCI 1320 that is based on GC-DCI 2_3 Type A, and which supports configuration of partial resource sounding. Enhanced GC-DCI Type A provides a block 1322, 1324, 1326 for N UEs or groups of UEs. Each block 1322, 1324, 1326 corresponds to a set of carrier frequencies and includes one or more different TPC entries 1330. An SRS request identifier 1328 is used to select a CC set (carrier frequency). The TPC entries 1330 includes an entry for each of the corresponding CC sets.

(120) In one aspect, partial frequency sounding commands (PFS commands 1332) may be configured for the corresponding CC sets. The PFS commands 1332 may include a bitmap or index to a preconfigured table. The bitmap or index may identify the SRS resources to be used by a UE for partial sounding. In one example, M bits are added to each block 1322, 1324, 1326 for every CC, for a total of $M \times (\text{number of CCs in the CC set})$ bits that are used to indicate the sub frequency resources to be sounded within each CC. In another example, a 4-bit index may be used to index a table that includes 16 entries covering up to 4 CCs and 4 sub-frequency resources per block 1322, 1324, 1326. FIG. 14 illustrates a second example of enhanced GC-DCI 1420 that is based on GC-DCI 2_3 Type B, and which supports configuration of partial resource sounding. Enhanced GC-DCI Type B provides a block 1422, 1424, 1426 for N carrier frequencies. Each block 1422, 1424,

1426 corresponds to a single carrier frequency and includes a TPC entry **1430** for the corresponding carrier frequency. An SRS request identifier **1438** is used to select an individual CC (carrier frequency).

(121) In one aspect, partial frequency sounding commands (PFS commands **1432**) may be configured for the corresponding CC sets. The PFS commands **1432** may include a bitmap or index to a preconfigured table. The bitmap or index may identify the SRS resources to be used by a UE for partial sounding. In one example, M bits are added to each block **1422**, **1424**, **1426** for every CC, for a total of $M \times (\text{number of CCs in the CC set})$ bits that are used to indicate the sub frequency resources to be sounded within each CC. In another example, a 4-bit index may be used to index a table that includes 16 entries covering up to 4 CCs and 4 sub-frequency resources per block **1422**, **1424**, **1426**.

(122) FIG. **15** illustrates a third example of enhanced GC-DCI **1500** that is based on GC-DCI 2_3 Type A, and which supports configuration of partial resource sounding. Enhanced GC-DCI **1500** provides a block **1502**, **1504**, **1506** for N UEs or groups of UEs. Each block **1502**, **1504**, **1506** corresponds to a set of carrier frequencies and includes one or more different TPC entries **1512**. An SRS request identifier **1510** is used to select a CC set (carrier frequency). The TPC entries **1512** includes an entry for each of the corresponding CC sets.

(123) In one aspect the enhanced GC-DCI **1500** includes a field a CC set identifier **1508** used to select an individual CC (carrier frequency) that can be used to trigger multiple CCs simultaneously. In another aspect, partial frequency sounding commands (PFS commands **1332**) may be configured for the corresponding CC sets. The PFS commands **1332** may include a bitmap or index to a preconfigured table. The bitmap or index may identify the SRS resources to be used by a UE for partial sounding. In one example, M bits are added to each block **1522**, **1524**, **1526** for every CC, for a total of $M \times (\text{number of CCs in the CC set})$ bits that are used to indicate the sub frequency resources to be sounded within each CC. In another example, a 4-bit index may be used to index a table that includes 16 entries covering up to 4 CCs and 4 sub-frequency resources per block **1522**, **1524**, **1526**.

(124) In another aspect, the SRS request identifier **1510** provides flexible SRS triggering for different SRS usages, including antenna switching for beam management, codebook, etc.

(125) In another aspect, the PFS command field **1332**, **1432**, **1514** in enhanced GC-DCI carries a bitmap that can be used to select sub frequency resources to be sounded. The bitmap may directly identify the sub frequency resources. In the hopping example illustrated in FIG. **9**, the first RB group **912** (also referred to as the first hop) includes 4 sub-hops which can be allocated among 4 UEs. In this example, each sub-hop includes a single RB **902**, **904**, **906**, **908**, although one or more sub-hops may include more than one of the RBs **902**, **904**, **906**, **908**. A first UE may be configured with a bitmap with the value '1000' indicating that the first UE can use the first sub-hop for sounding and that the remaining 3 sub-hops are unavailable or unused for sounding by the first UE.

(126) In the non-hopping example illustrated in FIG. **10**, the RBs allocated for SRS by the base station may be divided into four RB groups **1002**, **1004**, **1006**, **1008**, and each of the RB groups **1002**, **1004**, **1006**, **1008** can be allocated for the use of different UEs. Each of the RB groups **1002**, **1004**, **1006**, **1008** includes 4 RBs, although one or more of the RB groups **1002**, **1004**, **1006**, **1008** may include more than four RBs. A first UE may be configured with a bitmap with the value '1000' indicating that the first UE can use the sub-band that includes the first RB group **1002** for sounding and that the remaining 3 RB groups **1004**, **1006**, **1008** are unavailable or unused for sounding by the first UE.

(127) In some implementations, a bitmap provided by the PFS command field **1332**, **1432**, **1514** in enhanced GC-DCI carries is valid for all hops or transmissions. In some implementations, cycling within the hops can be indicated by RRC signaling. In one example, the bitmap may be cycled in a sequence such as {1000.fwdarw.0100.fwdarw.0010.fwdarw.0001.fwdarw.1000 . . . }.

(128) In some instances, the bitmap may indicate that more than one sub-resource (sub-hop or sub-

band) may be sounded. For example, a bitmap with the value '1001' indicates that the first UE can use the first and last sub-hops or the first and last sub-bands for sounding. The bitmap can support non-contiguous resource sounding including, for example, when the bit map has the value '1010' or '0101' for example.

(129) In one aspect, the PFS command field **1332**, **1432**, **1514** in enhanced GC-DCI carries an index that indicates an entry in an RRC-configured table. The entry in the table may identify which sub-frequency resource (sub-hop or sub-band) is assigned to the corresponding UE to be sounded. In the 4-UE, 4 sub-frequency resources examples illustrated in FIGS. **9**, **10** and **12**, a 2-bit index is sufficient to indicate one of the four possible entries of pre-configured table. The use of an index can provide benefits over bitmaps when the sub-frequency resources are unequally split, when larger numbers of UEs are to be configured, and when it is desirable to increase the number of properties that can be referenced in the preconfigured table. The size of the index may be calculated as $\log_2$ (number of configurations to be supported), where the number of configurations may correspond to the number of resources for sounding use. The use of an index requires that the table be preconfigured, typically by RRC signaling.

(130) FIG. **16** is a conceptual diagram illustrating an example of a hardware implementation for an exemplary UE **1600** employing a processing system **1614**. For example, the UE **1600** may be any of the UEs or scheduled entities illustrated in any one or more of FIGS. **1**, **3** and/or **4**.

(131) The UE **1600** may be implemented with a processing system **1614** that includes one or more processors **1604**. Examples of processors **1604** include microprocessors, microcontrollers, digital signal processors (DSPs), field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. In various examples, the UE **1600** may be configured to perform any one or more of the functions described herein. That is, the processor **1604**, as utilized in a UE **1600**, may be used to implement any one or more of the processes described below in connection with FIG. **16**.

(132) The processor **1604** may in some instances be implemented via a baseband or modem chip and in other implementations, the processor **1604** may itself comprise a number of devices distinct and different from a baseband or modem chip (e.g., in such scenarios as may work in concert to achieve aspects discussed herein). And as mentioned above, various hardware arrangements and components outside of a baseband modem processor can be used in implementations, including RF-chains, power amplifiers, modulators, buffers, interleavers, adders/summers, etc.

(133) In this example, the processing system **1614** may be implemented with a bus architecture, represented generally by the bus **1602**. The bus **1602** may include any number of interconnecting buses and bridges depending on the specific application of the processing system **1614** and the overall design constraints. The bus **1602** communicatively couples together various circuits including one or more processors (represented generally by the processor **1604**), a memory **1605**, and computer-readable media (represented generally by the computer-readable medium **1606**). The bus **1602** may also link various other circuits such as timing sources, peripherals, voltage regulators, and power management circuits, which are well known in the art, and therefore, will not be described any further. A bus interface **1608** provides an interface between the bus **1602** and a transceiver **1610**. The transceiver **1610** provides a means for communicating with various other apparatus over a transmission medium (e.g., air interface). In some examples, the transceiver **1610** may include a phase-shifter **1616** for digital and/or analog beamforming via one or more antenna array(s) **1630**. A user interface **1612** (e.g., keypad, display, speaker, microphone, joystick) may also be provided.

(134) The processor **1604** is responsible for managing the bus **1602** and general processing, including the execution of software stored on the computer-readable medium **1606**. The software, when executed by the processor **1604**, causes the processing system **1614** to perform the various functions described below for any particular apparatus. The computer-readable medium **1606** and

the memory **1605** may also be used for storing data that is manipulated by the processor **1604** when executing software.

(135) One or more processors **1604** in the processing system may execute software. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, etc., whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. The software may reside on a computer-readable medium **1606**.

(136) The computer-readable medium **1606** may be a non-transitory computer-readable medium. A non-transitory computer-readable medium includes, by way of example, a magnetic storage device (e.g., hard disk, floppy disk, magnetic strip), an optical disk (e.g., a compact disc (CD) or a digital versatile disc (DVD)), a smart card, a flash memory device (e.g., a card, a stick, or a key drive), a random access memory (RAM), a read only memory (ROM), a programmable ROM (PROM), an erasable PROM (EPROM), an electrically erasable PROM (EEPROM), a register, a removable disk, and any other suitable medium for storing software and/or instructions that may be accessed and read by a computer. The computer-readable medium **1606** may reside in the processing system **1614**, external to the processing system **1614**, or distributed across multiple entities including the processing system **1614**. The computer-readable medium **1606** may be embodied in a computer program product. In some examples, the computer-readable medium **1606** may be part of the memory **1605**. By way of example, a computer program product may include a computer-readable medium in packaging materials. Those skilled in the art will recognize how best to implement the described functionality presented throughout this disclosure depending on the particular application and the overall design constraints imposed on the overall system.

(137) In some aspects of the disclosure, the processor **1604** may include circuitry configured for various functions. For example, the processor **1604** may include communication and processing circuitry **1642**, configured to communicate with one or more radio access network (RAN) entities, such as one or more base stations (e.g., gNBs), and/or one or more scheduling entities. In some examples, the communication and processing circuitry **1642** may include one or more hardware components that provide the physical structure that performs processes related to wireless communication (e.g., signal reception and/or signal transmission) and signal processing (e.g., processing a received signal and/or processing a signal for transmission).

(138) In some examples, the communication and processing circuitry **1642** may be configured to receive and process downlink beamformed signals at a mmWave frequency or a sub-6 GHz frequency via the transceiver **1610** and the antenna array **1630** (e.g., using the phase-shifter **1616**). In addition, the communication and processing circuitry **1642** may be configured to generate and transmit uplink beamformed signals at a mmWave frequency or a sub-6 GHz frequency via the transceiver **1610** and antenna array **1630** (e.g., using the phase-shifter **1616**).

(139) The communication and processing circuitry **1642** may further be configured to execute communication and processing software **1652** stored in the computer-readable medium **1606** to implement one or more of the functions described herein.

(140) The processor **1604** may further include SRS manager circuitry **1644**, configured to process a partial frequency sounding configuration **1620** defined for one or more UEs. In some examples, the partial frequency sounding configuration **1620** may include an identification of certain resources to be used for transmitting SRS.

(141) The SRS manager circuitry **1644** may further be configured to execute SRS manager software **1654** stored in the computer-readable medium **1606** to implement one or more of the functions described herein.

(142) In one configuration, the UE **1600** includes means for performing the various functions and processes described in relation to FIG. **18**. In one aspect, the aforementioned means may be the processor **1604** shown in FIG. **16** configured to perform the functions recited by the

aforementioned means. In another aspect, the aforementioned means may be a circuit or any apparatus configured to perform the functions recited by the aforementioned means.

(143) Of course, in the above examples, the circuitry included in the processor **1604** is merely provided as an example, and other means for carrying out the described functions may be included within various aspects of the present disclosure, including but not limited to the instructions stored in the computer-readable storage medium **1606**, or any other suitable apparatus or means described in any one of the FIGS. **1, 3, 4**, and/or **7-10**, and utilizing, for example, the processes and/or algorithms described herein in relation to FIG. **18**.

(144) FIG. **17** is a conceptual diagram illustrating an example of a hardware implementation for an exemplary RAN entity **1700** employing a processing system **1714**. For example, the RAN entity **1700** may correspond to any of the base stations (e.g., gNBs), TRPs (e.g., combined TRP and base station in a RRH configuration), or other scheduling entities illustrated in any one or more of FIGS. **1, 3, 4**, and/or **7-10**.

(145) In accordance with various aspects of the disclosure, an element, or any portion of an element, or any combination of elements may be implemented with a processing system **1714** that includes one or more processors **1704**. The processing system **1714** may be substantially the same as the processing system **1614** illustrated in FIG. **16**, including a bus interface **1708**, a bus **1702**, memory **1705**, a processor **1704**, and a computer-readable medium **1706**. Furthermore, the RAN entity **1700** may include an optional user interface **1712** and a transceiver **1710** substantially similar to those described above in FIG. **16**. In some examples, the transceiver **1710** may include a phase-shifter **1716** for digital and/or analog beamforming via one or more antenna array(s) **1730**. The processor **1704**, as utilized in a RAN entity **1700**, may be used to implement any one or more of the processes described below.

(146) In some aspects of the disclosure, the processor **1704** may include circuitry configured for various functions. For example, the processor **1704** may include resource assignment and scheduling circuitry **1742**, configured to generate, schedule, and modify a resource assignment or grant of time-frequency resources (e.g., a set of one or more resource elements). For example, the resource assignment and scheduling circuitry **1742** may schedule time-frequency resources within a plurality of time division duplex (TDD) and/or frequency division duplex (FDD) subframes, slots, and/or mini-slots to carry user data traffic and/or control information to and/or from multiple UEs.

(147) In some examples, the resource assignment and scheduling circuitry **1742** may be configured to schedule resources for the transmission of an SRS configuration for a multi-TRP SRS resource set to a user equipment (UE) in communication with the RAN entity **1700**. The resource assignment and scheduling circuitry **1742** may further be configured to schedule resources for the reception of one or more SRSs from the UE in accordance with the SRS configuration. The resource assignment and scheduling circuitry **1742** may further be configured to schedule resources for the transmission of DCI triggering the SRS configuration or a MAC-CE activating the SRS configuration. In addition, the resource assignment and scheduling circuitry **1742** may further be configured to schedule resources for the transmission of a TPC command to the UE. The resource assignment and scheduling circuitry **1742** may further be configured to execute resource assignment and scheduling software **1752** stored in the computer-readable medium **1706** to implement one or more of the functions described herein.

(148) The processor **1704** may further include communication and processing circuitry **1744**, configured to communicate with the UE. In some examples, the communication and processing circuitry **1744** may include one or more hardware components that provide the physical structure that performs processes related to wireless communication (e.g., signal reception and/or signal transmission) and signal processing (e.g., processing a received signal and/or processing a signal for transmission).

(149) In some examples, the communication and processing circuitry **1744** may be configured to receive and process uplink beamformed signals at a mmWave frequency or a sub-6 GHz frequency

via the transceiver **1710** and the antenna array **1730** (e.g., using the phase-shifter **1716**). In addition, the communication and processing circuitry **1744** may be configured to generate and transmit uplink beamformed signals at a mmWave frequency or a sub-6 GHz frequency via the transceiver **1710** and antenna array **1730** (e.g., using the phase-shifter **1716**).

(150) The communication and processing circuitry **1744** may further be configured to transmit an SRS configuration **1720** for the multi-TRP SRS resource set to the UE. The multi-TRP SRS resource set may include a plurality of SRS resources associated with the multiple TRPs. The multiple TRPs may be separate base stations or RRHs of a same base station. In the former example, the RAN entity **1700** may correspond to one of the base stations. In the latter example, the RAN entity **1700** may be coupled to each of the multiple TRPs. In some examples, the multiple TRPs may be co-located or positioned at different geographic locations and may belong to the same cell or different cells.

(151) The multi-TRP SRS resource set may be, for example, an aperiodic SRS resource set, periodic SRS resource set, or semi-persistent SRS resource set. In some examples, the SRS configuration may be transmitted within a RRC message. In some examples, the RRC message may include a single RRC configuration (e.g., a single RRC configuration information element (IE)) including the SRS configuration. In other examples, the RRC message may include multiple RRC configurations (e.g., multiple RRC configuration IEs), each including one of a plurality of TRP SRS configurations, each associated with one of the TRPs, where the plurality of TRP SRS configurations collectively form the SRS configuration. The communication and processing circuitry **1744** may further be configured to store the SRS configuration **1720** within, for example, the memory **1705**.

(152) The communication and processing circuitry **1744** may further be configured to transmit an activation/trigger for the multi-TRP SRS resource set. For example, the communication and processing circuitry **1744** may transmit DCI triggering an aperiodic multi-TRP SRS resource set or a MAC-CE activating a semi-persistent multi-TRP SRS resource set. In some examples, the communication and processing circuitry **1744** may be configured to transmit DCI including a codepoint triggering the SRS configuration (e.g., when the SRS configuration includes a single RRC configuration) or a codepoint triggering each of the plurality of TRP SRS configurations (e.g., when the SRS configuration includes multiple RRC configurations).

(153) The communication and processing circuitry **1744** may further be configured to transmit at least one closed-loop TPC command to the UE. The TPC command may be for either PUSCH transmissions or SRS transmissions. For example, a TPC command for SRS may be received in DCI format 2_3. In some examples, the TPC command is associated with one of the TRPs. In other examples, the TPC command may be applicable to each of the TRPs. In addition, the communication and processing circuitry **1744** may be configured to transmit a timing advance command to the UE. The timing advance command indicates a timing advance to be applied by the UE to uplink signals transmitted to the TRPs within a TAG. Here, at least one of the TRPs in the TAG is associated with (e.g., part of) the RAN entity **1700**. The communication and processing circuitry **1744** may further be configured to receive one or more SRSs in accordance with the SRS configuration. The communication and processing circuitry **1744** may further be configured to execute communication and processing software **1754** stored in the computer-readable medium **1706** to implement one or more of the functions described herein.

(154) The processor **1704** may further include SRS manager circuitry **1746**, configured to configure the SRS configuration for the multi-TRP SRS resource set for the UE. In some examples, the SRS configuration may include a common set of SRS resource set parameters for the multiple TRPs. For example, the multi-TRP SRS resource set may include one or more SRS resources that each include the same (common) SRS resource parameters (e.g., transmission comb structure, port(s), number of symbols, repetition, etc.). In other examples, the SRS configuration may include a respective set of SRS resource set parameters for each of the multiple TRPs. For example, the

SRS configuration can include a plurality of TRP SRS configurations, each associated with a respective TRP, where each TRP SRS configuration includes different respective SRS resource parameters. Each TRP SRS configuration may configure one or more SRS resources associated with a particular TRP. Thus, each TRP SRS configuration may include the SRS resource set parameters configuring the particular SRS resources for that TRP.

(155) In some examples, when each of the TRPs belongs to the same TAG and the QCL association for SRS resources configured for each of the TRPs is the same, the SRS manager circuitry **1746** can configure the SRS resources associated with the multiple TRPs to be located within consecutive symbols of a slot. In some examples, when two of the TRPs may belong to different TAGs or the QCL association for SRS resources configured for two of the TRPs is different, the SRS manager circuitry **1746** may be configured to determine a gap length of one or more symbols to be applied between the SRS resources associated with each of the TRPs. For example, the SRS manager circuitry **1746** may be configured to determine the gap length to be applied between a first set of SRS resources associated with a first TRP and a second set of SRS resources associated with a second TRP.

(156) In some examples, when the TRPs belong to different TAGs, the SRS manager circuitry **1746** may set the gap length to accommodate a maximum timing advance difference that may occur between the TAGs. In other examples, when the QCL association between the SRS resources associated with each of the TRPs is different, the SRS manager circuitry **1746** may set the gap length based on a capability of the UE or as a common gap length configured for all UEs. In some examples, the SRS manager circuitry **1746** may transmit an indication of the gap length as part of the SRS resource configuration (e.g., within the SRS configuration **1720**) or separately from the SRS configuration via the communication and processing circuitry **1744** and transceiver **1710**.

(157) The SRS manager circuitry **1746** may further receive and process each of the SRSs received from the UE to obtain respective measurements (e.g., RSRP, RSRQ, etc.) of each of the received SRSs for uplink beam management. The SRS manager circuitry **1746** may further be configured to execute SRS manager software **1756** stored in the computer-readable medium **1706** to implement one or more of the functions described herein.

(158) In one configuration, the RAN entity **1700** includes means for performing the various functions and processes described in relation to FIG. **19**. In one aspect, the aforementioned means may be the processor **1704** shown in FIG. **17** configured to perform the functions recited by the aforementioned means. In another aspect, the aforementioned means may be a circuit or any apparatus configured to perform the functions recited by the aforementioned means.

(159) Of course, in the above examples, the circuitry included in the processor **1704** is merely provided as an example, and other means for carrying out the described functions may be included within various aspects of the present disclosure, including but not limited to the instructions stored in the computer-readable storage medium **1706**, or any other suitable apparatus or means described in any one of the FIG. **1**, **3** or **4** and utilizing, for example, the processes and/or algorithms described herein in relation to FIG. **19**.

(160) FIG. **18** is a flow chart **1800** illustrating an example of a method for a UE to utilize a partial frequency sounding configuration according to some aspects. As described below, some or all illustrated features may be omitted in a particular implementation within the scope of the present disclosure, and some illustrated features may not be required for implementation of all aspects. In some examples, the method may be performed by the UE **1600**, as described above and illustrated in FIG. **16**, by a processor or processing system, or by any suitable means for carrying out the described functions.

(161) At block **1802**, the UE may receive DCI that includes a partial frequency sounding configuration for the UE. At block **1804**, the UE may transmit a SRS in one or more resources configured for SRS transmission in a bandwidth used by the UE. The one or more resources may be identified by the partial frequency sounding configuration. At block **1802**, the UE may refrain

from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE.

(162) In one example, the partial frequency sounding configuration includes a bitmap that includes a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.

(163) In one example, the UE may identify the one or more resources configured for SRS transmission by the UE in a preconfigured table using an index provided in the partial frequency sounding configuration.

(164) In one example, the downlink control information includes a block corresponding to the UE, the block including the partial frequency sounding configuration for the UE.

(165) In certain examples, the downlink control information has a format that is based on GC-DCI defined for use in a 5G NR network. The format of the downlink control information may be based on GC-DCI 2_3 Type A and may include a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources. In some examples, the downlink control information further includes a component carrier index used to identify the component carrier set associated with the partial frequency sounding configuration in a list of component carrier sets. The format of the downlink control information may be based on GC-DCI 2_3 Type B and may include a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.

(166) In one example, the one or more resources configured for SRS transmission include a plurality of contiguous frequency resources within the bandwidth used by the UE.

(167) In one example, the one or more resources configured for SRS transmission include a plurality of non-contiguous frequency resources within the bandwidth used by the UE.

(168) In one example, the one or more resources configured for SRS transmission include frequency resources provisioned at different times.

(169) FIG. **19** is a flow chart **1900** illustrating an example of a method for a RAN entity to configure a partial frequency sounding configuration for a plurality of UEs according to some aspects. As described below, some or all illustrated features may be omitted in a particular implementation within the scope of the present disclosure, and some illustrated features may not be required for implementation of all aspects. In some examples, the method may be performed by the RAN entity **1700**, as described above and illustrated in FIG. **17**, by a processor or processing system, or by any suitable means for carrying out the described functions.

(170) At block **1902**, the RAN entity may configure a SRS configuration to be used by at least one UE. The SRS configuration may include, provision or define a plurality of frequency resources. At block **1904**, the RAN entity may generate partial frequency sounding configurations for a plurality of UEs. At block **1906**, the RAN entity may transmit the partial frequency sounding configurations in DCI. Each partial frequency sounding configuration can define one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS.

(171) In one example, each partial frequency sounding configuration includes a bitmap that includes a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources to be used by the corresponding UE for transmitting the SRS.

(172) In one example, the RAN entity may configure each of the plurality of UEs with a table that defines the plurality of frequency resources and provide an index in each partial frequency sounding configuration. The index may identify the one or more resources in the corresponding preconfigured table.

(173) In one example, the RAN entity may transmit each of the partial frequency sounding configurations in a block of the DCI that is identified with a corresponding UE.

(174) In certain examples, the RAN entity may transmit the partial frequency sounding

configurations in downlink control information that has a format that is based on GC-DCI defined for use in a 5G NR network. The format of the downlink control information may be based on GC-DCI 2_3 Type A and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table or a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources. The DCI may include a component carrier index used to identify the component carrier set associated with the partial frequency sounding configuration in a list of component carrier sets. The list of component carrier sets may be included in the preconfigured table that identifies the one or more resources. The format of the DCI may be based on GC-DCI 2_3 Type B and may include a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table or a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.

(175) In one example, the one or more resources configured for SRS transmission include a plurality of contiguous frequency resources within the bandwidth used by the UE.

(176) In one example, the one or more resources configured for SRS transmission include a plurality of non-contiguous frequency resources within the bandwidth used by the UE.

(177) In one example, the one or more resources configured for SRS transmission include frequency resources provisioned at different times.

(178) The following provides an overview of examples of the present disclosure.

(179) Several aspects of a wireless communication network have been presented with reference to an exemplary implementation. As those skilled in the art will readily appreciate, various aspects described throughout this disclosure may be extended to other telecommunication systems, network architectures and communication standards.

(180) By way of example, various aspects may be implemented within other systems defined by 3GPP, such as Long-Term Evolution (LTE), the Evolved Packet System (EPS), the Universal Mobile Telecommunication System (UMTS), and/or the Global System for Mobile (GSM). Various aspects may also be extended to systems defined by the 3rd Generation Partnership Project 2 (3GPP2), such as CDMA2000 and/or Evolution-Data Optimized (EV-DO). Other examples may be implemented within systems employing IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Ultra-Wideband (UWB), Bluetooth, and/or other suitable systems. The actual telecommunication standard, network architecture, and/or communication standard employed will depend on the specific application and the overall design constraints imposed on the system.

(181) Within the present disclosure, the word “exemplary” is used to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term “coupled” is used herein to refer to the direct or indirect coupling between two objects. For example, if object A physically touches object B, and object B touches object C, then objects A and C may still be considered coupled to one another even if they do not directly physically touch each other. For instance, a first object may be coupled to a second object even though the first object is never directly physically in contact with the second object. The terms “circuit” and “circuitry” are used broadly, and intended to include both hardware implementations of electrical devices and conductors that, when connected and configured, enable the performance of the functions described in the present disclosure, without limitation as to the type of electronic circuits, as well as software implementations of information and instructions that, when executed by a processor, enable the performance of the functions described in the present disclosure.

(182) One or more of the components, steps, features and/or functions illustrated in FIGS. 1-14 may be rearranged and/or combined into a single component, step, feature or function or embodied in several components, steps, or functions. Additional elements, components, steps, and/or

functions may also be added without departing from novel features disclosed herein. The apparatus, devices, and/or components illustrated in FIGS. 1, 3, 4, and/or 7-12 may be configured to perform one or more of the methods, features, or steps described herein. The novel algorithms described herein may also be efficiently implemented in software and/or embedded in hardware.

(183) It is to be understood that the specific order or hierarchy of steps in the methods disclosed is an illustration of exemplary processes. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the methods may be rearranged. The accompanying method claims present elements of the various steps in a sample order and are not meant to be limited to the specific order or hierarchy presented unless specifically recited therein.

(184) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. A phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a; b; c; a and b; a and c; b and c; and a, b, and c. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. A method for wireless communication at a user equipment (UE) in a wireless communication network, the method comprising: receiving downlink control information that includes a partial frequency sounding configuration for the UE; transmitting a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration; and refraining from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE; wherein the downlink control information has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type A and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
2. The method of claim 1, wherein the downlink control information includes a component carrier index used to identify the component carrier set associated with the partial frequency sounding configuration in a list of component carrier sets.
3. A method for wireless communication at a user equipment (UE) in a wireless communication network, the method comprising: receiving downlink control information that includes a partial frequency sounding configuration for the UE; transmitting a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration; and refraining from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE; wherein the downlink control information has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-

- DCI 2_3 Type B and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
4. A method for wireless communication at a radio access network (RAN) entity in a wireless communication network, the method comprising: configuring a sounding reference signal (SRS) configuration to be used by at least one user equipment (UE), the SRS configuration comprising a plurality of frequency resources; generating partial frequency sounding configurations for a plurality of UEs; transmitting the partial frequency sounding configurations in downlink control information, wherein each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS; and transmitting the partial frequency sounding configurations in downlink control information that has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type A and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
5. The method of claim 4, wherein the downlink control information includes a component carrier index used to indicate the component carrier set associated with the partial frequency sounding configuration in a list of component carrier sets.
6. A method for wireless communication at a radio access network (RAN) entity in a wireless communication network, the method comprising: configuring a sounding reference signal (SRS) configuration to be used by at least one user equipment (UE), the SRS configuration comprising a plurality of frequency resources; generating partial frequency sounding configurations for a plurality of UEs; transmitting the partial frequency sounding configurations in downlink control information, wherein each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS; and transmitting the partial frequency sounding configurations in downlink control information that has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type B and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
7. A user equipment (UE) configured for wireless communication, comprising: one or more processors; and one or more memories coupled to the one or more processors, the one more processors and the one or more memories configured to: receive downlink control information that includes a partial frequency sounding configuration for the UE; transmit a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration; and refrain from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE; wherein the downlink control information has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type A and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
8. A user equipment (UE) configured for wireless communication, comprising: one or more processors; and one or more memories coupled to the one or more processors, the one more processors and the one or more memories configured to: receive downlink control information that

includes a partial frequency sounding configuration for the UE; transmit a sounding reference signal (SRS) in one or more resources configured for SRS transmission in a bandwidth used by the UE, the one or more resources being identified by the partial frequency sounding configuration; and refrain from transmitting in a portion of a set of resources configured for SRS transmission in the bandwidth used by the UE; wherein the downlink control information has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type B and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.

9. A radio access network (RAN) entity configured for wireless communication, comprising: one or more processors; and one or more memories coupled to the one or more processors, the one or more processors and the one or more memories configured to: configure a sounding reference signal (SRS) configuration to be used by at least one user equipment (UE), the SRS configuration comprising a plurality of frequency resources; generate partial frequency sounding configurations for a plurality of UEs; transmit the partial frequency sounding configurations in downlink control information, wherein each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS; and transmit the partial frequency sounding configurations in downlink control information that has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type A and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.

10. A radio access network (RAN) entity configured for wireless communication, comprising: one or more processors; and one or more memories coupled to the one or more processors, the one or more processors and the one or more memories configured to: configure a sounding reference signal (SRS) configuration to be used by at least one user equipment (UE), the SRS configuration comprising a plurality of frequency resources; generate partial frequency sounding configurations for a plurality of UEs; transmit the partial frequency sounding configurations in downlink control information, wherein each partial frequency sounding configuration defines one or more resources in the plurality of frequency resources to be used by a corresponding UE for transmitting an SRS; and transmit the partial frequency sounding configurations in downlink control information that has a format that is based on group common downlink control information (GC-DCI) defined for use in a fifth-generation new radio (5G NR) network; wherein the format of the downlink control information is based on GC-DCI 2_3 Type B and includes a partial frequency sounding field that includes a resource index used to identify the one or more resources in a preconfigured table, or includes a bitmap having a plurality of bits, each bit in the bitmap being mapped to a subset of the one or more resources.
